DEEP LEARNING THE NANOSCALE: A PHYSICS-INFORMED 3D UNET FOR SOLVING THE THERMOPLASMONIC INVERSE PROBLEM

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Kushagra Sinha

ABSTRACT

Quantitative phase imaging (QPI) techniques, such as thermal imaging by quadriwave shearing interferometry (TQSI), have emerged as powerful non-invasive tools for mapping temperature distributions around optically heated plasmonic nanostructures. However, these measurements inherently project the three-dimensional thermal field onto a two-dimensional optical phase map through path integration of the thermo-optic response $\Delta n(T)$ along the optical axis. Recovering the full volumetric temperature distribution $T(x, y, z)$ from a single 2D phase measurement constitutes a severely ill-posed inverse problem, particularly in stratified media where thermal conductivity varies discontinuously across material boundaries.

In this work, we develop a complete computational framework bridging analytical physics and deep learning to solve this thermoplasmonic inverse problem. First, we construct a rigorous analytical forward model based on the thermal Green's function for a three-layer stratified medium (substrate–liquid–superstrate), derived using the method of image charges with exact reflection and transmission coefficients. This forward solver, implemented as GPU-accelerated FFT-based spatial convolutions, generates synthetic paired datasets of 2D phase maps and their corresponding 3D temperature tensors for diverse nanoparticle topologies, materials, and illumination profiles.

Second, we design a physics-informed 3D U-Net architecture (`PhysicsGuidedInversionUNet`) that performs 2D-to-3D extrusion through a learned spatial fusion mechanism. The network is regularized by a custom loss function (`FastAdaptiveLagrangianLoss`) that enforces the steady-state Laplace equation $\nabla^2 T = 0$ in source-free regions using differentiable 3D finite-difference kernels, while simultaneously constraining boundary conditions, gradient fidelity, and source-plane localization. We additionally develop an automated experimental data-cleaning pipeline for off-axis digital holographic interferometry, incorporating super-Gaussian sideband isolation, radial-profile-based camera harmonic detection, and DBSCAN-clustered notch filtering. Preliminary results demonstrate the network's ability to reconstruct volumetric thermal fields with accurate z-axis source localization from single 2D phase inputs.

LIST OF SYMBOLS AND ABBREVIATIONS

Symbol	Description
σ_{abs}	Absorption Cross-Section
σ_{ext}	Extinction Cross-Section
σ_{sca}	Scattering Cross-Section
Q_{abs}	Absorption Efficiency
$\Delta\phi$	Optical Phase Difference
κ	Thermal Conductivity
$G(\vec{r}, \vec{r}_0)$	Thermal Green's Function
β_i	Thermo-Optic Coefficient (order i)
Δn	Refractive Index Change
ΔT	Temperature Rise Above Ambient
$R_{\text{top}}, R_{\text{bot}}$	Reflection Coefficients (Image Charges)
T_{ij}	Transmission Coefficients
W	Convergence Parameter ($R_{\text{top}} \cdot R_{\text{bot}}$)
I	Laser Irradiance (W/m^2)
∇^2	Laplacian Operator
$\mathcal{F}$	Fourier Transform

Abbreviation	Full Form
LSPR	Localized Surface Plasmon Resonance
TQSI	Thermal imaging by Quantitative Phase Shearing Interferometry
QPI	Quantitative Phase Imaging
PINN	Physics-Informed Neural Network
UNet	U-shaped Convolutional Network
FFT	Fast Fourier Transform
OPD	Optical Path Difference
TDMS	Technical Data Management Streaming
MAD	Median Absolute Deviation
DBSCAN	Density-Based Spatial Clustering
AMP	Automatic Mixed Precision

LIST OF FIGURES

1.1 Schematic of the thermoplasmonic heating process. An incident laser excites a Localized Surface Plasmon Resonance (LSPR) in a metallic nanoparticle. Non-radiative relaxation of the plasmon leads to Joule heating, which diffuses into the surrounding medium, creating a 3D thermal landscape $T(x, y, z)$ and a corresponding refractive index gradient. 3

1.2 Schematic of the dual-beam interferometric thermometry apparatus at the IISERB Light Lab. A 785 nm pump laser directly targets the sample (S) via objective O_3 to excite plasmonic resonances. Concurrently, a 457 nm probe laser is divided by a fiber splitter into an objective path (probing the thermal phase shift through O_2 and O_3) and a reference path. Both paths recombine at beam splitter BS_1 to project an interferogram onto the camera, which is mathematically processed to extract the exact phase variations induced by the heated nanostructures. 5

1.3 Overview of the project workflow. The analytical Green's function model (Forward Model) generates synthetic training data for complex stratified media. Experimental TQSI data is heavily cleaned via an automated pipeline. Finally, the Physics-Informed 3D UNet ingests the 2D phase maps and outputs the full 3D temperature field, constrained by the Laplacian loss. 8

2.1 The quasistatic picture of a localized surface plasmon. The incident electric field displaces the conduction electron cloud relative to the positively charged ion lattice, inducing a dipole moment. This creates a uniform internal electric field and a classic dipolar distortion of the external field lines. 13

3.1 Block diagram of the FFT-based forward solver pipeline, illustrating the generation of the power map, padding, Fourier convolution, and nonlinear phase residual computation. 24

3.2 Forward solver simulation for a **Single Isolated Nanoparticle**. *Specifications:* Gold (Au) nanosphere, radius $a = 50$ nm; Ambient medium: Homogeneous Water ($\kappa_{eff} = 0.60$ W/m·K, $n_{med} = 1.33$); Particle refractive index $n_s = 0.54 + 2.23j$; Illumination: Uniform plane wave at $\lambda = 532$ nm, Intensity $I_0 = 100.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 2.0894 \times 10^{-2} \mu$m^2. The middle panel visualizes the finite-sized thermal Green's function kernel $G(x, y, 0)$ bounded at $r = a$, preventing a central singularity. The bottom panel plots the strict $1/r$ thermal decay profile dropping smoothly from the surface peak ΔT along the 1D cross-section at $y = 0$. 25

3.3 Forward solver simulation for a **Square Nanoparticle Array under Uniform Illumination**. *Specifications:* Gold (Au) nanospheres, radius $a = 40$ nm; Array dimensions: 15×15 particles with lattice pitch $p = 600$ nm; Ambient medium: Homogeneous Water ($\kappa_{eff} = 0.60$ W/m·K, $n_{med} = 1.33$); Particle refractive index $n_s = 0.54 + 2.23j$; Illumination: Uniform plane wave at $\lambda = 532$ nm, Constant Intensity $I_0 = 50.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 1.8289 \times 10^{-2} \mu$m^2. The discrete source matrix $Q(x, y)$ mirrors the flat-top spatial envelope, resulting in a broad collective heating profile superimposed with uniform, highly localized intra-particle thermal spikes approaching a ΔT of 13 K. 26

3.4 Forward solver simulation for a **Hexagonal Lattice under Gaussian Illumination**. *Specifications:* Silver (Ag) nanospheres, radius $a = 30$ nm; Hexagonal close-packed (HCP) 2D lattice bounded by a 10 μm circular aperture with nearest-neighbor pitch $p = 450$ nm; Ambient medium: Isopropanol (IPA, $\kappa_{eff} = 0.135$ W/m·K, $n_{med} = 1.37$); Particle refractive index $n_s = 0.05 + 3.42j$; Illumination: Gaussian beam profile at $\lambda = 532$ nm, beam waist $w_0 = 8.0\mu$m, Peak Intensity $I_0 = 50.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 3.1174 \times 10^{-4} \mu$m^2. The lower thermal conductivity of the IPA medium strongly restricts heat dissipation compared to water, heavily amplifying the macroscopic collective thermal envelope across the dense lattice. 27

3.5 Detailed steps of the TDMS data cleaning procedure. The 12-panel plot illustrates the full diagnostic pipeline, detecting harmonic noise and filtering it out in the Fourier domain. 30

3.6 Comparison of the phase map before and after the automated cleaning procedure, showcasing the removal of periodic artifacts while preserving the true sample response. 31

3.7 Gallery of generated synthetic topologies showcasing Bravais lattice, geometric ring, amorphous random distribution, and multi-cluster arrays. . . 32

3.8 Placeholder for a table listing the materials database, detailing respective thermal conductivities (κ), refractive indices (n), and thermo-optic polynomials (β). 33

4.1 Detailed illustration of the complete end-to-end pipeline. Multiple heat sources residing in the source plane generate a 3D thermal field. The optical accumulation yields the 2D phase map $\Delta\phi(x, y)$. The architecture processes this alongside the strictly structured 3D physical tensors (κ and β_{3D}) using spatial extrusion to fuse dimensions, subsequently predicting the spatial shape map and dynamic magnitude independently to resolve the z-location ambiguity. 36
4.2 Neuronic map of the Physics-Guided Inversion UNet, explicitly detailing the feature connections, skip layers, and the crucial bifurcation into the magnitude scaler and the spatial shape sigmoid map. 37
4.3 Decomposition of the FastAdaptiveLagrangianLoss into purely data-driven, structural, and physics-based penalty terms. 39
4.4 Training metrics over 100 epochs, demonstrating rapid convergence of the data loss and stabilization of the physics Laplacian loss. 41

5.1 Forward model validation across four topological test cases. From left to right: input phase map, ground truth temperature, PINN prediction, and relative error map. The cross-sectional profiles in the lower panels demonstrate exact agreement between analytical and boundary-adjusted solutions. 44
5.2 Qualitative PINN reconstruction results. Slices at varying z-depths show the input phase map (left), ground truth temperature (middle), and the PINN predicted 3D output (right). Error maps show deviations primarily confined to sub-pixel high-gradient regions. 45
5.3 Z-localization accuracy of the PINN. The softmax-driven Z-locator branch successfully isolates the source plane with minimal error, overcoming the ill-posed nature of depth recovery from 2D blurred phase maps. 45
5.4 1D temperature profiles (cross-cuts) along the x and z axes. The PINN perfectly tracks the $1/r$ thermal decay and respects the asymmetric boundary conditions induced by multi-layer environments. 46
5.5 Training convergence curves showing the evolution of the data loss, physics (Laplacian) loss, and Z-locator loss over epochs. 46
5.6 Experimental data processing pipeline. Raw holograms are filtered in the Fourier domain to remove camera harmonics, yielding high signal-to-noise ratio phase maps for thermal reconstruction. 47

CONTENTS

Certificate i

Academic Integrity and Copyright Disclaimer ii

Acknowledgement iii

Abstract iv

List of Symbols and Abbreviations v

List of Figures ix

1. Introduction and Motivation 1
- 1.1 The Age of Nanoscale Heat Management 1
- 1.2 Localized Surface Plasmon Resonance 2
- 1.3 Thermal Imaging of Plasmonic Nanostructures 3
- 1.4 The Inverse Problem: From 2D Phase to 3D Temperature 6
- 1.5 Physics-Informed Deep Learning: A New Paradigm 7
- 1.6 Scope and Contributions of This Report 9

2. Theoretical Foundations of Thermoplasmonics 10
- 2.1 Electrodynamics of Metallic Nanoparticles 10
  - 2.1.1 The Drude Model 10
  - 2.1.2 The Quasistatic Approximation and Polarizability of a Nanosphere 11
  - 2.1.3 Optical Cross-Sections from the Polarizability 13
  - 2.1.4 Beyond the Quasistatic Limit: Full Mie Theory 14
  - 2.1.5 Mie-Gans Theory for Nanorods 15
- 2.2 Photothermal Heating 16
  - 2.2.1 From Light Absorption to Heat Generation 16
  - 2.2.2 The Steady-State Heat Diffusion Equation 16
- 2.3 Green's Function for a Three-Layer Stratified Medium 17
- 2.4 The Thermo-Optic Effect and Phase Measurement 19

3. Forward Modeling and Experimental Data Processing 21
3.1 The FFT-Based Forward Solver 21
3.1.1 Spatial Convolution in Fourier Space 21
3.1.2 Forward Solver Algorithm 22
3.1.3 3D Volumetric Extension 23
3.1.4 Representative Forward Simulations for PINN Training 24
3.2 Experimental Data Processing Pipeline 27
3.2.1 Off-Axis Digital Holographic Interferometry 27
3.2.2 Sideband Isolation via Super-Gaussian Windowing 28
3.2.3 Automated Camera Harmonic Noise Detection 28
3.2.4 Phase Extraction and OPD Conversion 31
3.3 Synthetic Dataset Generation for Neural Network Training 32
3.3.1 Ultra-Diverse Topology Engine 32
3.3.2 Physical Parameter Randomization 32
3.3.3 Adaptive Temperature Rescaling 33

4. Physics-Informed Deep Learning for the Inverse Problem 34
4.1 The Failure of Standard Analytical Inversion 34
4.1.1 Linear Tikhonov Deconvolution 34
4.1.2 Iterative Nonlinear Inversion 35
4.1.3 Why a Neural Network? 35
4.2 Network Architecture and Pipeline: The Physics-Guided Inversion UNet 36
4.2.1 Pipeline Overview and Constraints 36
4.2.2 Code Usage Example 36
4.2.3 Design Philosophy 37
4.2.4 The 2D Phase Encoder 37
4.2.5 The 3D Physical Prior Encoder 37
4.2.6 SpatialExtrusionFusion: Bridging 2D to 3D 37
4.2.7 The 3D U-Net Decoder 38
4.2.8 The Magnitude-Aware Output 38
4.3 The FastAdaptiveLagrangianLoss: Physics as Regularization 39
4.3.1 The 3D Finite-Difference Laplacian Kernel 39
4.3.2 Loss Decomposition 39
4.3.3 Why the Laplacian Weight is 0.05 41
4.4 The Z-Guided ThermalUNet: An Alternative Architecture 41
4.5 Training Methodology 41
4.5.1 Mixed-Precision Training with Gradient Accumulation 42
4.5.2 Optimization Schedule 42

5. Results, Benchmarking, and Conclusion . . . 43
5.1 Forward Model Validation . . . 43
5.1.1 Benchmark Against Published Data . . . 43
5.1.2 Multi-Layer System Verification . . . 43
5.2 PINN 3D Reconstruction Results . . . 44
5.2.1 Qualitative Assessment . . . 44
5.2.2 Z-Axis Source Localization . . . 45
5.2.3 Temperature Magnitude Recovery . . . 45
5.2.4 Cross-Cut Profile Analysis . . . 45
5.3 Training Convergence Analysis . . . 46
5.4 Experimental Data Processing Results . . . 46
5.5 Discussion . . . 47
5.5.1 Strengths of the Approach . . . 47
5.5.2 Current Limitations . . . 47
5.6 Conclusion . . . 47
5.7 Future Directions . . . 48

Appendices . . . 49
I Code Snippets: Physics-Informed Loss Function . . . 50
I.1 The FastAdaptiveLagrangianLoss Implementation . . . 50
I.2 The 3D Laplacian Kernel Construction . . . 54
I.3 Materials Database . . . 55
I.4 Training Hyperparameters . . . 56

Bibliography . . . 60

1. INTRODUCTION AND MOTIVATION

1.1 The Age of Nanoscale Heat Management

Heat is ubiquitous in physical processes, often viewed as an unavoidable byproduct of energy conversion. However, when the generation and flow of heat are precisely controlled at the nanoscale, it ceases to be a mere waste product and instead becomes a powerful tool. The ability to manipulate temperature profiles on length scales comparable to or smaller than the wavelength of visible light has given rise to the vibrant field of thermoplasmonics Govorov and Richardson [2007], Brongersma, Halas, and Nordlander [2015].

In macroscopic systems, heat diffuses over large volumes, leading to slow thermal dynamics and broad spatial temperature gradients. At the nanoscale, however, the physics of heat transfer undergoes a paradigm shift. Nanoparticles can act as highly localized, rapidly tunable heat sources. This extraordinary spatiotemporal control has profound implications across multiple disciplines. In medicine, for instance, functionalized gold nanoparticles can be directed to attach to tumor cells. Upon laser irradiation, they act as nanoscopic stoves, destroying the cancer cells through localized hyperthermia without damaging the surrounding healthy tissue. In chemistry, nanoscale heat sources drive localized catalytic reactions (nanochemistry), shifting reaction equilibria and increasing yields at specific catalytic sites. In optics and information technology, highly confined thermal fields are utilized in photothermal imaging, optical trapping, and heat-assisted magnetic recording (HAMR), which promises to vastly increase the density of data storage.

To harness these applications effectively, we must possess the ability not only to generate nanoscale heat but also to accurately measure and model it in three-dimensional space. The complexity lies in the fact that conventional thermometry fails at these length scales. Thermocouples and infrared cameras are restricted by their physical size and the diffraction limit of mid-infrared light, respectively, preventing them from resolving temperature distributions around single nanoparticles. This project addresses the challenge of mapping these invisible, nanoscopic thermal landscapes by bridging the gap between cutting-edge optical interferometry and modern deep learning techniques.

1.2 Localized Surface Plasmon Resonance

To understand how a nanoparticle can serve as an efficient, localized heat source, we must explore its interaction with electromagnetic radiation. When a metallic nanoparticle—such as gold or silver—is illuminated by light, the oscillating electric field of the incident electromagnetic wave exerts a force on the free conduction electrons in the metal. This causes the electron cloud to collectively oscillate relative to the positively charged ionic lattice.

When the frequency of the incident light matches the natural frequency of the electron cloud oscillating against the restoring force of the lattice, a resonance condition is met. This phenomenon is known as Localized Surface Plasmon Resonance (LSPR) Maier [2007]. Unlike propagating surface plasmons on flat metal-dielectric interfaces, LSPR is confined to the finite volume of the nanoparticle, meaning no phase-matching techniques (like prisms or gratings) are required to excite it.

Qualitatively, the LSPR induces two primary effects. First, it leads to a strong enhancement of the local electric field near the nanoparticle surface—the foundation of techniques like Surface-Enhanced Raman Spectroscopy (SERS). Second, it increases the particle's ability to absorb and scatter light. The optical response of the nanoparticle can be quantified by its absorption cross-section, σ_{abs}, which represents the effective area over which the particle captures electromagnetic energy.

When the particle is much smaller than the incident wavelength ($R \ll \lambda$), the phase of the electromagnetic field is essentially constant over the particle volume. We can therefore treat the electromagnetic problem electrostatically, a regime known as the quasistatic approximation Maier [2007]. By solving Laplace's equation for a sphere in a uniform electric field, we obtain a dipole moment. This dipole moment defines the particle's polarizability α (the full derivation is provided in Chapter 2):

$$\alpha = 4\pi R^3 \frac{\varepsilon_m - \varepsilon_d}{\varepsilon_m + 2\varepsilon_d} \tag{1.1}$$

where R is the particle radius, ε_m is the complex dielectric function of the metal, and ε_d is the dielectric constant of the surrounding medium. The absorption cross-section σ_{abs} follows directly from the polarizability:

$$\sigma_{\text{abs}} = k\Im(\alpha) \tag{1.2}$$

where $k = 2\pi\sqrt{\varepsilon_d}/\lambda$ is the wavevector in the host medium. Resonance occurs when the denominator of the polarizability is minimized. This Fröhlich resonance condition, $\text{Re}(\varepsilon_m) \approx -2\varepsilon_d$, leads to a sharp maximum in σ_{abs}.

Once the light is absorbed, the energy is stored in the excited electron gas (hot electrons). Within picoseconds, these hot electrons scatter with phonons, transferring

their kinetic energy to the crystal lattice and rapidly raising the internal temperature of the nanoparticle. The heat is then dissipated into the surrounding medium via thermal conduction. **Assumption:** Under continuous-wave illumination at steady state, the total heat power Q (in Watts) generated by a nanoparticle is directly proportional to the incident irradiance I (in W/m^2) and its absorption cross-section:

$$Q = \sigma_{\text{abs}} I \tag{1.3}$$

Equation 1.3 is a cornerstone of thermoplasmonics Baffou [2017]. It dictates that by tuning the shape, size, and material of the nanoparticle to maximize σ_{abs} at the laser wavelength, we can engineer highly efficient, sub-wavelength heat sources.

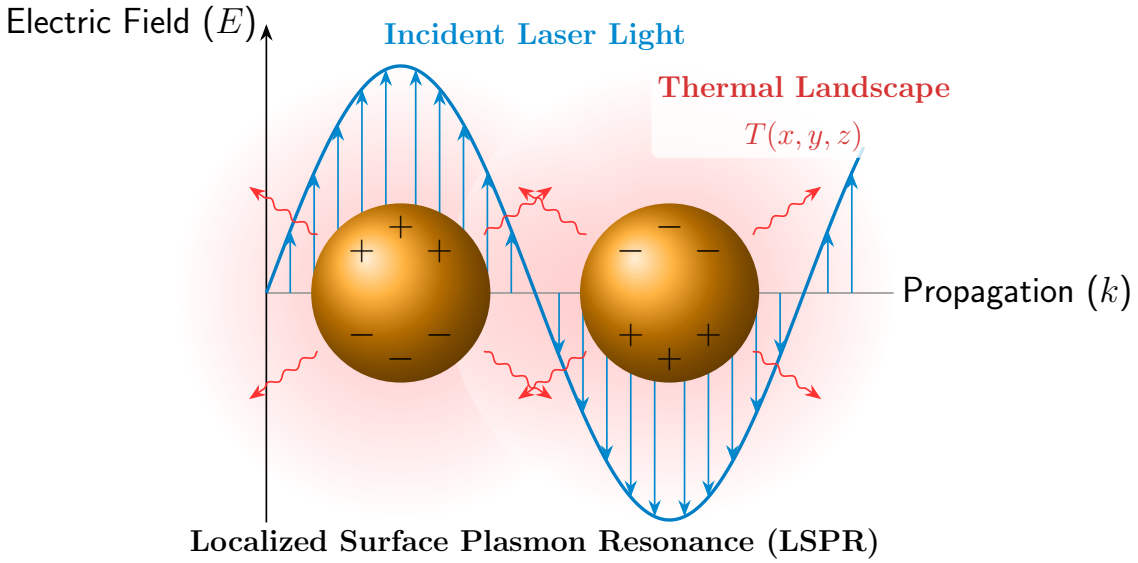


Fig. 1.1: Schematic of the thermoplasmonic heating process. An incident laser excites a Localized Surface Plasmon Resonance (LSPR) in a metallic nanoparticle. Non-radiative relaxation of the plasmon leads to Joule heating, which diffuses into the surrounding medium, creating a 3D thermal landscape $T(x, y, z)$ and a corresponding refractive index gradient.

1.3 Thermal Imaging of Plasmonic Nanostructures

While generating heat is straightforward, measuring the resulting temperature distribution $T(x, y, z)$ is highly challenging. As previously mentioned, direct mechanical probing or standard IR imaging is impossible at the nanoscale. To circumvent these limitations, researchers rely on indirect optical methods. One of the most powerful techniques developed in recent years is based on quantitative phase imaging (QPI), specifically Thermal Quadriwave Lateral Shearing Interferometry (TQSI) introduced by Baffou and colleagues Baffou, Bon, Savatier, Polleux, Zhu, Merlin, Rigneault, and Monneret [2012], Baffou, Ureña, Berto, Monneret, Quidant, and Rigneault [2014], Bon, Maucort, Wattellier, and

Monneret [2009].

The physical principle underlying TQSI is the thermo-optic effect. When the nanoparticle heats the surrounding medium (e.g., water or a polymer), the local density of the medium changes, which in turn alters its refractive index n. For small temperature variations, this relationship is linear:

$$\Delta n(x, y, z) = \frac{dn}{dT} \Delta T(x, y, z) \tag{1.4}$$

where dn/dT is the thermo-optic coefficient of the medium (typically on the order of -10^{-4} K^{-1} for liquids).

Instead of trying to measure the temperature directly, QPI measures the optical footprint left by this refractive index gradient. A planar wavefront of a probe light beam (independent of the heating laser) is transmitted through the sample. As the light propagates through the heated, inhomogeneous medium, it accumulates a localized optical path difference, leading to a distortion in the wavefront phase.

Using a modified diffraction grating (a modified Hartmann mask) and a standard CCD camera, TQSI can extract this 2D phase map $\Delta\phi(x, y)$ with high sensitivity. The measured phase shift is the integral of the refractive index variation along the optical axis (z-axis):

$$\Delta\phi(x, y) = \frac{2\pi}{\lambda_{\text{probe}}} \int_{-\infty}^{\infty} \Delta n(x, y, z)\, dz = \frac{2\pi}{\lambda_{\text{probe}}} \frac{dn}{dT} \int_{-\infty}^{\infty} \Delta T(x, y, z)\, dz \tag{1.5}$$

Equation 1.5 reveals a key limitation of optical thermometry. We have successfully translated a sub-wavelength thermal measurement into an optical phase measurement. However, because the probe beam integrates the signal along its path, all depth-resolved information (the z-dependence of the temperature) is inherently lost. We are left with a 2D projection of a 3D physical reality.

To practically realize these quantitative phase measurements and extract accurate thermal maps, a highly customized interferometric setup is required. Figure 1.2 illustrates the dual-beam thermometry apparatus developed and utilized at the IISERB Light Lab IISER Bhopal [2024]. This setup isolates the heating mechanics from the phase-probing mechanics.

The localized heating is driven by a 785 nm pump laser. The pump beam is shaped by a series of lenses (L_8, L_9) and a spatial aperture, steered by mirrors (M_3, M_4), and reflected by a dichroic mirror (DM_1) into the primary objective (O_3). This objective focuses the intense light onto the sample (S), directly exciting the LSPR of the plasmonic nanostructures and inducing localized Joule heating.

Simultaneously, a separate 457 nm probe laser maps the resulting refractive index gradient. The probe beam passes through lenses L_1 and L_2, and is focused by a coupling

lens (L_c) into a fiber coupler. A 1×2 fiber splitter divides the coherent light into two distinct paths: an objective path and a reference path. The objective beam exits O_1, traverses a complex conditioning relay (including lenses L_3–L_7 and a wedge prism for fine angular alignment), and is focused onto the sample by objective O_2. Crucially, as this beam propagates through the thermally expanded medium surrounding the nanoparticles, its wavefront accumulates a spatial phase shift $\Delta\phi(x, y)$ in accordance with Equation 1.5. After traversing the sample, the probe beam is collected by O_3 and, because of its specific wavelength, is transmitted directly through the dichroic mirror DM_1.

In the lower arm of the apparatus, the unperturbed reference beam exits a collimator and passes through a polarizing beam splitter (PBS) and a half-wave plate (HWP), allowing precise matching of intensity and polarization state with the objective beam. The reference and objective wavefronts are finally recombined at beam splitter BS_1. This superposition creates a high-contrast interferogram, which is spectrally filtered to block any scattered 785 nm pump light and captured by a high-resolution camera. Subsequent computational data analysis demodulates this interference pattern to retrieve the exact phase shift map, allowing researchers to accurately infer the thermal landscape generated by the nanostructures.

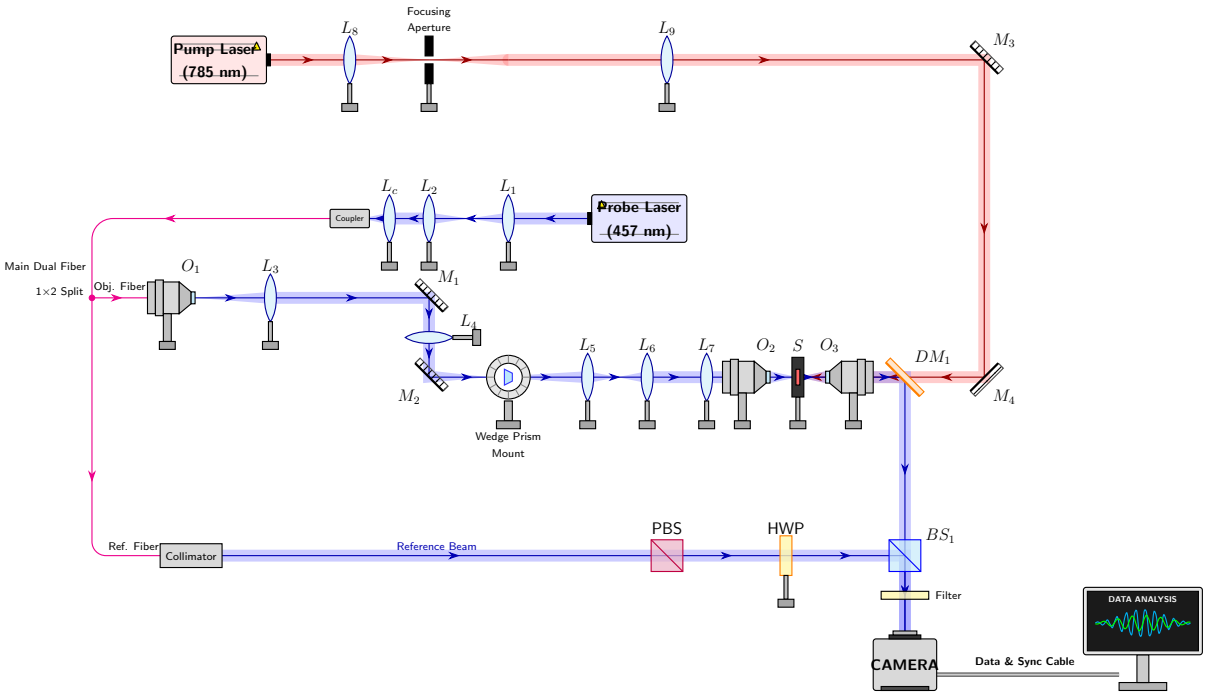


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1.4 The Inverse Problem: From 2D Phase to 3D Temperature

We have arrived at the central mathematical and computational challenge of this report: the inverse problem. The *forward problem* is conceptually straightforward: given a set of heat sources Q_i at known positions, we solve the steady-state heat equation to find $\Delta T(x, y, z)$, and then integrate along z to simulate the measured phase map $\Delta\phi(x, y)$.

The *inverse problem* asks the opposite: given an experimentally measured 2D phase map $\Delta\phi(x, y)$, can we reconstruct the full 3D temperature distribution $\Delta T(x, y, z)$ and determine the exact heat power Q generated by the nanoparticles?

Mathematically, recovering a 3D volumetric field from a single 2D projection is ill-posed. According to Jacques Hadamard (1902), a problem is *well-posed* if a solution exists, the solution is unique, and the solution's behavior changes continuously with the initial conditions Hadamard [1902]. Our inverse problem violates the uniqueness condition—infinite different 3D temperature configurations could theoretically project onto the exact same 2D phase map. Furthermore, measurement noise in the phase map is amplified during inversion, violating the stability condition. In principle, tomographic methods utilizing multiple projections could break the z-ambiguity, but TQSI provides only a single projection—making the problem fundamentally underdetermined.

The standard approach in the literature is to attempt a deconvolution. Assuming the system is linear, the phase map can be expressed as a convolution between the 2D heat source density $q(x, y)$ and a point spread function (the phase response of a single point source). By applying Tikhonov regularization Tikhonov, Goncharsky, Stepanov, and Yagola [1995] in Fourier space, one can stabilize the inversion and retrieve the 2D power density. However, this classical approach has severe limitations:

1. It yields only the 2D projection of the heat source density, not the 3D temperature field.

2. It assumes an infinite homogeneous medium to simplify the Green's function inversion. In reality, thermoplasmonic experiments occur in stratified media (e.g., a glass substrate, a water layer, and a superstrate). In such asymmetric environments, the thermal Green's function possesses a complex z-dependent structure. Simple 1/r inversion becomes ambiguous and physically inaccurate.

To understand the thermal dynamics in realistic, asymmetric experimental setups, we need an inversion framework capable of ingesting 2D projection data and outputting a full 3D, physics-compliant thermal profile.

1.5 Physics-Informed Deep Learning: A New Paradigm

In recent years, deep learning has advanced inverse problem-solving in imaging and computational physics. The naive approach to our problem would be a purely data-driven supervised learning model: train a Convolutional Neural Network (CNN) on thousands of paired examples of 2D phase maps (input) and their corresponding 3D temperature profiles (output).

However, purely data-driven models are "black boxes." They learn statistical correlations rather than physical laws. Consequently, they struggle to generalize outside their training distribution, require massive amounts of labeled data, and often produce predictions that violate the fundamental laws of thermodynamics (e.g., predicting heat flow from cold to hot, or non-zero Laplacian fields in source-free regions).

To overcome these hurdles, we turn to Physics-Informed Neural Networks (PINNs), pioneered by Raissi, Perdikaris, and Karniadakis in 2019 Raissi, Perdikaris, and Karniadakis [2019], Karniadakis, Kevrekidis, Lu, Perdikaris, Wang, and Yang [2021]. The key insight of PINNs is direct: rather than relying solely on data, we embed the governing partial differential equations (PDEs) directly into the neural network's loss function.

Assumption: Assuming steady-state heat conduction in source-free media, the temperature field outside the nanoparticles must obey Laplace's equation:

$$\nabla^2(\Delta T) = 0 \quad \text{(in source-free regions)} \tag{1.6}$$

In a PINN framework, the neural network acts as a parameterized ansatz for the temperature field, $T_\theta(x, y, z)$, where θ represents the network weights. During training, the loss function $\mathcal{L}$ is formulated as a weighted sum of two components:

$$\mathcal{L} = \mathcal{L}_{\text{data}} + \lambda \mathcal{L}_{\text{physics}} \tag{1.7}$$

Here, $\mathcal{L}_{\text{data}}$ ensures the 2D projection of the network's 3D output matches the experimentally measured phase map. Simultaneously, $\mathcal{L}_{\text{physics}}$ penalizes the residual of the PDE (i.e., it forces $\nabla^2 T_\theta \to 0$ everywhere outside the particles). Using automatic differentiation, we can compute exact spatial derivatives of the neural network with respect to its input coordinates, without relying on error-prone finite difference grids Cai, Mao, Wang, Yin, and Karniadakis [2021].

By injecting strong physical priors into the learning process, the PINN regularizes the severely ill-posed inverse problem. The physics loss constrains the hypothesis space, shrinking the set of mathematically possible solutions down to only those that are physically realizable.

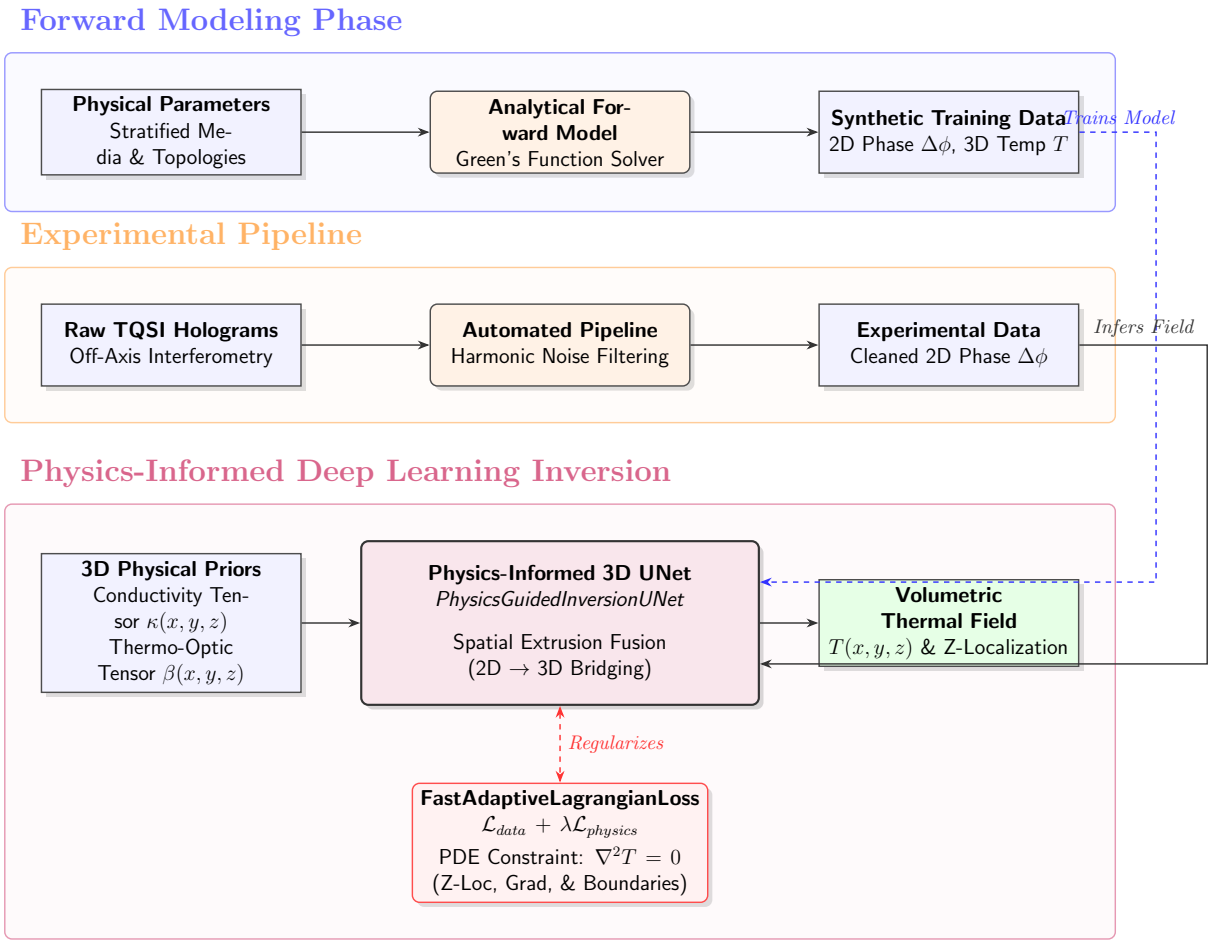


Fig. 1.3: Overview of the project workflow. The analytical Green's function model (Forward Model) generates synthetic training data for complex stratified media. Experimental TQSI data is heavily cleaned via an automated pipeline. Finally, the Physics-Informed 3D UNet ingests the 2D phase maps and outputs the full 3D temperature field, constrained by the Laplacian loss.

1.6 Scope and Contributions of This Report

The primary objective of this project is to construct, train, and validate a comprehensive physics-informed deep learning framework capable of solving the 3D thermoplasmonic inverse problem in realistic, multi-layered environments. We transition from classical Tikhonov regularization to a modern, neural-network-driven paradigm, providing researchers with a robust tool to "see" 3D nanoscale heat.

The specific contributions of this work are fourfold:

1. **Analytical 3-Layer Green's Function Forward Solver:** We derive and implement a highly optimized, semi-analytical forward solver for the heat equation in a glass/water/air stratified medium. This solver acts as the bedrock for generating high-fidelity synthetic data.

2. **Automated Experimental Data Pipeline:** We develop an automated, robust image processing pipeline to clean raw experimental TQSI phase maps, addressing critical issues like baseline phase tilt, high-frequency noise, and sensor artifacts.

3. **Physics-Informed 3D UNet Architecture:** We design a custom, asymmetric neural network architecture that takes 2D planar data as input and hallucinates a physically accurate 3D volume, utilizing skip connections and spatial upsampling to preserve high-frequency thermal features.

4. **Custom Lagrangian Loss with Laplacian PDE Enforcement:** We formulate a highly specialized composite loss function, enforcing Laplace's equation in source-free volumes and applying continuity constraints across the substrate/superstrate interfaces using exact automatic differentiation.

The remainder of this report is structured as follows. Chapter 2 provides the rigorous theoretical foundation of nanoscale heat transfer and optical phase integration. Chapter 3 details the computational implementation of our forward model and the intricacies of the data generation process. Chapter 4 introduces the architecture, hyperparameter tuning, and training dynamics of our proposed Physics-Informed Deep Learning framework. Finally, Chapter 5 presents the quantitative results, benchmarking the network's 3D thermal reconstructions against synthetic ground truth and validating its performance on real-world experimental data, followed by concluding remarks and avenues for future exploration.

2. THEORETICAL FOUNDATIONS OF THERMOPLASMONICS

2.1 Electrodynamics of Metallic Nanoparticles

2.1.1 The Drude Model

To understand how metallic nanoparticles interact with light, we must first describe the optical properties of the bulk metal itself. The simplest yet most physically intuitive approach is the Drude model.

Assumption: We treat the conduction electrons in the metal as a classical, non-interacting free-electron gas moving against a fixed background of positive ion cores. We explicitly neglect interband transitions (bound electron effects), which become important at shorter wavelengths.

In this classical picture, an electron of mass m_e and charge $-e$ subjected to a time-harmonic electric field $\mathbf{E}(t) = \mathbf{E}_0 e^{-i\omega t}$ experiences a driving force. The equation of motion is:

$$m_e \frac{\partial^2 \mathbf{x}}{\partial t^2} + m_e \gamma \frac{\partial \mathbf{x}}{\partial t} = -e\mathbf{E}_0 e^{-i\omega t}, \tag{2.1}$$

where $\gamma = 1/\tau$ is the collision frequency (or damping rate) representing electron scattering with phonons, defects, and other electrons, and τ is the mean relaxation time. Assuming a steady-state oscillatory solution $\mathbf{x}(t) = \mathbf{x}_0 e^{-i\omega t}$, we find:

$$\mathbf{x}_0 = \frac{e}{m_e(\omega^2 + i\gamma\omega)}\mathbf{E}_0. \tag{2.2}$$

The macroscopic polarization $\mathbf{P}$, defined as the dipole moment per unit volume, is $\mathbf{P} = -ne\mathbf{x}$, where n is the electron number density. From macroscopic electrodynamics, the dielectric function $\varepsilon(\omega)$ relates to the polarizability via $\mathbf{P} = \varepsilon_0(\varepsilon(\omega) - 1)\mathbf{E}$. Substituting our expression for $\mathbf{x}$, we arrive at the Drude dielectric function:

$$\varepsilon(\omega) = 1 - \frac{\omega_p^2}{\omega^2 + i\gamma\omega}, \tag{2.3}$$

where $\omega_p = \sqrt{ne^2/\varepsilon_0 m_e}$ is the bulk plasma frequency. At high frequencies ($\omega \gg \gamma$), the metal acts as a lossless, dispersive medium, and light cannot propagate if $\omega < \omega_p$ (giving

metals their characteristic reflective properties).

While beautifully simple, this model fails to capture the full picture for noble metals like gold and silver. Real metals exhibit interband transitions—for instance, in gold, d-band electrons are excited to the sp-conduction band at energies above ~ 2.4 eV ($\lambda \approx 516$ nm). This drastically alters the dielectric response. In practice, we rely on experimental dielectric data measured by Johnson and Christy Johnson and Christy [1972] to correctly model real metals in the visible and near-infrared regimes.

2.1.2 The Quasistatic Approximation and Polarizability of a Nanosphere

We now turn from bulk metals to nanoparticles. When a metallic nanosphere is illuminated by light, the free electrons in the metal are driven by the oscillating electric field, creating a collective excitation known as a localized surface plasmon resonance (LSPR). To capture this physics elegantly, we start with the quasistatic approximation.

Assumptions:

- The particle radius R is much smaller than the wavelength of incident light ($R \ll \lambda$). This allows us to neglect retardation effects, meaning the electric field is effectively uniform across the particle at any given instant.
- We assume all materials are non-magnetic ($\mu = \mu_0$).
- The media are isotropic, linear, and homogeneous.
- We adopt the time-harmonic convention $e^{-i\omega t}$ for all oscillating fields.

The problem reduces to solving an electrostatic boundary value problem, as derived by Maier Maier [2007] and Jackson Jackson [1999]. Consider a metallic sphere of radius R and complex dielectric function $\varepsilon(\omega)$, embedded in a non-absorbing host medium with a real dielectric constant ε_m. The system is subjected to a uniform static electric field $\mathbf{E} = E_0\hat{\mathbf{z}}$.

Because $R \ll \lambda$, we solve Laplace's equation for the electric potential, $\nabla^2\Phi = 0$. In spherical coordinates (r, θ, ϕ), azimuthal symmetry implies no ϕ-dependence. The general solution is a superposition of Legendre polynomials $P_l(\cos\theta)$:

$$\Phi(r,\theta) = \sum_{l=0}^{\infty} \left(A_l r^l + \frac{B_l}{r^{l+1}} \right) P_l(\cos\theta). \tag{2.4}$$

We apply standard electrostatic boundary conditions:

1. The potential must approach the applied field far from the particle: $\Phi_{\text{out}} \to -E_0 r\cos\theta$ as $r \to \infty$.

2. The potential must be finite at the center of the particle ($r = 0$).

3. At the sphere boundary $r = R$, the potential Φ (tangential $\mathbf{E}$) and the normal component of the displacement field $\varepsilon \frac{\partial \Phi}{\partial r}$ (normal $\mathbf{D}$) must be continuous.

Due to the form of the applied field (which goes as $\cos\theta = P_1(\cos\theta)$) and the orthogonality of Legendre polynomials, only the $l = 1$ terms survive. Inside the sphere, the $1/r^2$ term diverges, so $B_1^{\text{in}} = 0$. Outside, the r^1 term must match the applied field, so $A_1^{\text{out}} = -E_0$. Thus, we postulate solutions of the form:

$$\Phi_{\text{in}} = A_1^{\text{in}} r \cos\theta, \tag{2.5}$$

$$\Phi_{\text{out}} = -E_0 r \cos\theta + \frac{B_1^{\text{out}}}{r^2} \cos\theta. \tag{2.6}$$

Applying the continuity conditions at $r = R$:

$$A_1^{\text{in}} R = -E_0 R + \frac{B_1^{\text{out}}}{R^2}, \tag{2.7}$$

$$\varepsilon A_1^{\text{in}} = -\varepsilon_m E_0 - 2\varepsilon_m \frac{B_1^{\text{out}}}{R^3}. \tag{2.8}$$

Solving this system of equations for the coefficients yields:

$$A_1^{\text{in}} = -\frac{3\varepsilon_m}{\varepsilon + 2\varepsilon_m} E_0, \tag{2.9}$$

$$B_1^{\text{out}} = \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m} R^3 E_0. \tag{2.10}$$

Substituting these back, we obtain the internal and external potentials:

$$\Phi_{\text{in}} = -\frac{3\varepsilon_m}{\varepsilon + 2\varepsilon_m} E_0 r \cos\theta, \tag{2.11}$$

$$\Phi_{\text{out}} = -E_0 r \cos\theta + \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m} R^3 E_0 \frac{\cos\theta}{r^2}. \tag{2.12}$$

The physical insight here is profound. The external potential Φ_{out} is the superposition of the applied field and an induced dipole potential. The potential of an ideal dipole $\mathbf{p}$ aligned with the z-axis in a medium ε_m is $\Phi_{\text{dip}} = \frac{p\cos\theta}{4\pi\varepsilon_0\varepsilon_m r^2}$. By comparing this with the second term of Φ_{out}, we extract the induced dipole moment of the sphere:

$$p = 4\pi\varepsilon_0\varepsilon_m R^3 \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m} E_0. \tag{2.13}$$

We conventionally define the complex polarizability α through the relation $p = \varepsilon_0\varepsilon_m\alpha E_0$. This gives us the famous Clausius-Mossotti relation for the polarizability of a small sphere:

$$\alpha = 4\pi R^3 \frac{\varepsilon - \varepsilon_m}{\varepsilon + 2\varepsilon_m}. \tag{2.14}$$

Notice the denominator: $\varepsilon + 2\varepsilon_m$. The polarizability experiences a resonant enhancement when this denominator approaches zero. Because $\varepsilon = \mathrm{Re}(\varepsilon) + i\mathrm{Im}(\varepsilon)$, the exact condition is satisfied when:

$$\mathrm{Re}(\varepsilon(\omega)) = -2\varepsilon_m, \tag{2.15}$$

provided that $\mathrm{Im}(\varepsilon)$ is small. This is the Fröhlich condition. It is the defining requirement for a localized surface plasmon resonance. When met, the dipole moment is maximized, and the internal field of the nanoparticle is dramatically enhanced.

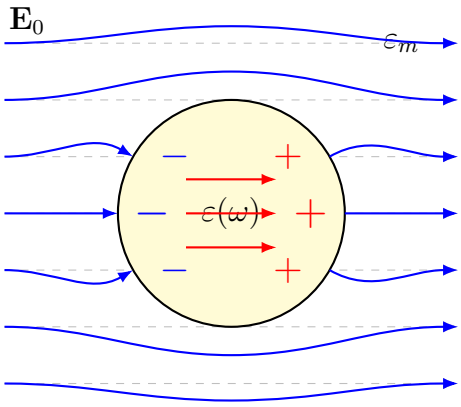


Fig. 2.1: The quasistatic picture of a localized surface plasmon. The incident electric field displaces the conduction electron cloud relative to the positively charged ion lattice, inducing a dipole moment. This creates a uniform internal electric field and a classic dipolar distortion of the external field lines.

2.1.3 Optical Cross-Sections from the Polarizability

Once we have the dipole moment of the nanoparticle, we can determine how it scatters and absorbs light. The time-varying dipole $\mathbf{p}(t)$ oscillates at the optical frequency ω, and therefore acts as a classic radiating antenna. From classical electrodynamics (the Larmor formula), the time-averaged power scattered by an oscillating dipole in a medium with wavenumber $k = 2\pi n_m/\lambda$ yields the scattering cross-section:

$$\sigma_{\mathrm{sca}} = \frac{k^4}{6\pi}|\alpha|^2. \tag{2.16}$$

Similarly, the optical theorem dictates that the total power extracted from the incident beam (the extinction cross-section) is proportional to the imaginary part of the forward scattering amplitude. For a dipole, this gives:

$$\sigma_{\mathrm{ext}} = k\mathrm{Im}(\alpha). \tag{2.17}$$

By conservation of energy, any light that is removed from the incident beam but not

scattered must be absorbed. Thus, the absorption cross-section is simply:

$$\sigma_{\text{abs}} = \sigma_{\text{ext}} - \sigma_{\text{sca}} = k\text{Im}(\alpha) - \frac{k^4}{6\pi}|\alpha|^2. \tag{2.18}$$

The scaling behavior here is the foundation of thermoplasmonics. For a small particle of radius R, the polarizability $|\alpha|$ scales with the particle volume, R^3. Therefore, $\sigma_{\text{abs}} \propto R^3$, whereas $\sigma_{\text{sca}} \propto R^6$. For very small particles (where $k^3|\alpha| \ll 1$), scattering falls off much faster than absorption. This means that small gold nanoparticles (e.g., $R < 20$ nm) are primarily *absorbers*, not scatterers. They convert virtually all intercepted optical energy directly into heat, making them ideal nano-heaters Maier [2007], Bohren and Huffman [1998].

2.1.4 Beyond the Quasistatic Limit: Full Mie Theory

The quasistatic approximation breaks down when particles exceed roughly 40 nm in diameter. Retardation effects kick in: the driving field is no longer uniform across the particle, leading to higher-order multipole excitations (quadrupoles, octupoles) and radiation damping. To calculate exact optical properties across all sizes, we must use the complete Mie theory, which solves Maxwell's equations analytically for spherical geometries Mie [1908], Bohren and Huffman [1998].

Mie theory relies on expanding the incident, scattered, and internal fields into vector spherical harmonics. The scattering problem is governed by the size parameter:

$$x = \frac{2\pi n_m R}{\lambda}, \tag{2.19}$$

and the relative refractive index $m = n_s/n_m$, where $n_s = \sqrt{\varepsilon}$ is the complex index of the sphere.

The scattering coefficients a_n (transverse magnetic modes) and b_n (transverse electric modes) are determined by the boundary conditions at the sphere surface. They are expressed using the Riccati-Bessel functions $\psi_n(z)$ and $\xi_n(z)$:

$$a_n = \frac{m\psi_n(mx)\psi_n'(x) - \psi_n(x)\psi_n'(mx)}{m\psi_n(mx)\xi_n'(x) - \xi_n(x)\psi_n'(mx)}, \tag{2.20}$$

$$b_n = \frac{\psi_n(mx)\psi_n'(x) - m\psi_n(x)\psi_n'(mx)}{\psi_n(mx)\xi_n'(x) - m\xi_n(x)\psi_n'(mx)}. \tag{2.21}$$

The exact solution requires summing over all mode numbers n from 1 to infinity. Numerically, the series converges rapidly. In our exact solver implementation, we truncate the series at the rigorous limit:

$$N_{\text{max}} = \text{round}\left(|x| + 4|x|^{1/3} + 2\right). \tag{2.22}$$

From the coefficients a_n and b_n, we calculate the dimensionless efficiency factors for extinction and scattering:

$$Q_{\text{ext}} = \frac{2}{x^2} \sum_{n=1}^{N_{\max}} (2n+1)\text{Re}(a_n + b_n), \tag{2.23}$$

$$Q_{\text{sca}} = \frac{2}{x^2} \sum_{n=1}^{N_{\max}} (2n+1)(|a_n|^2 + |b_n|^2). \tag{2.24}$$

The absorption efficiency is $Q_{\text{abs}} = Q_{\text{ext}} - Q_{\text{sca}}$. The absolute cross-sections are obtained by multiplying the efficiencies by the geometric cross-section πR^2. Importantly, taking the limit $x \to 0$ in Mie theory rigorously recovers the dipole polarizability of the quasistatic approximation.

2.1.5 Mie-Gans Theory for Nanorods

While Mie theory exactly solves the electrodynamics for perfect spheres, researchers often use anisotropic particles like gold nanorods to tune the plasmon resonance into the near-infrared biological window. The exact solution for finite cylinders requires numerical methods, but we can accurately approximate a nanorod as a prolate ellipsoid with length L and width W. This allows us to use the analytical Mie-Gans theory.

For an ellipsoid, the polarizability splits along the longitudinal (L) and transverse (T) axes. The static geometric depolarization factors are:

$$L_L = \frac{1-e^2}{e^2}\left[\frac{1}{2e}\ln\left(\frac{1+e}{1-e}\right) - 1\right], \tag{2.25}$$

$$L_T = \frac{1-L_L}{2}, \tag{2.26}$$

where $e = \sqrt{1-(W/L)^2}$ is the eccentricity.

To capture the redshift and broadening observed in larger nanorods, we apply Kuwata's dynamic depolarization correction Kuwata, Tamaru, Esumi, and Miyano [2003]:

$$\tilde{L}_L = L_L + A_L\varepsilon_m x_L^2 + B_L\varepsilon_m^2 x_L^4, \tag{2.27}$$

$$\tilde{L}_T = L_T + A_T\varepsilon_m x_T^2 + B_T\varepsilon_m^2 x_T^4, \tag{2.28}$$

where $x_{L,T} = \pi n_m(L,W)/(2\lambda)$, and $A_{L,T}, B_{L,T}$ are phenomenological geometry-dependent parameters.

The SI polarizability along each axis $j \in \{L, T\}$ for a particle of volume V becomes:

$$\alpha_{\text{stat},j} = V\frac{\varepsilon - \varepsilon_m}{\varepsilon_m + \tilde{L}_j(\varepsilon - \varepsilon_m)}. \tag{2.29}$$

To conserve energy properly, we must also apply the optical theorem radiation damping correction Link and El-Sayed [1999]:

$$\alpha_j = \frac{\alpha_{\text{stat},j}}{1 - i\frac{k^3}{6\pi}\alpha_{\text{stat},j}}. \tag{2.30}$$

Averaging over all random orientations yields the bulk cross-sections. This dynamic Mie-Gans approach accurately reproduces both the resonance shift and the radiation-broadened linewidth of real nanorods without requiring computationally heavy full-wave simulations.

2.2 Photothermal Heating

2.2.1 From Light Absorption to Heat Generation

The primary goal of thermoplasmonics is to utilize the large absorption cross-section of metallic nanoparticles to generate highly localized heat. The absorbed optical power by a single nanoparticle is simply:

$$Q = \sigma_{\text{abs}} I, \tag{2.31}$$

where I is the incident irradiance (W/m^2).

Assumption: We assume instantaneous thermalization.

The physical reason this assumption is valid relates to a cascade of energy transfer processes occurring on ultrafast timescales. First, the incident photons excite a collective plasmon resonance. This plasmon decays within tens of femtoseconds into a non-thermal distribution of hot electrons via Landau damping. The hot electrons rapidly thermalize among themselves via electron-electron scattering ($\sim$ 100 fs). They then transfer their energy to the crystalline lattice through electron-phonon coupling ($\sim 1 - 5$ ps). Finally, the heated lattice conducts thermal energy to the surrounding fluid on a timescale of $\sim$ 100 ps. Because our continuous-wave measurements occur over milliseconds to seconds, treating the nanoparticle as an instantaneous steady-state heat source is an extremely safe simplification Baffou, Berto, Ureña, Quidant, Monneret, Polleux, and Rigneault [2013], Baffou, Girard, and Quidant [2010a].

2.2.2 The Steady-State Heat Diffusion Equation

Assumption: We operate in the steady state ($\partial T/\partial t = 0$).

Once heat is deposited in the nanoparticle, it diffuses outward into the surrounding medium. The temperature distribution $T(\mathbf{r})$ is governed by the steady-state heat diffusion equation (Poisson equation):

$$\kappa \nabla^2 T(\mathbf{r}) = -q(\mathbf{r}), \tag{2.32}$$

where κ is the thermal conductivity of the medium and $q(\mathbf{r})$ is the volumetric heat source density.

For a spherical nanoparticle of radius R in an infinite homogeneous medium, the solution for the temperature increase $\Delta T = T - T_\infty$ outside the particle ($r \geq R$) takes the form of the free-space Green's function:

$$\Delta T(r) = \frac{Q}{4\pi\kappa r}. \tag{2.33}$$

The maximum temperature occurs at the particle surface ($r = R$):

$$\Delta T_{\max} = \frac{Q}{4\pi\kappa R}. \tag{2.34}$$

Assumption: For distances $r \gg R$, we invoke the point-source approximation. The exact geometry of the nanoparticle becomes irrelevant, and it acts purely as a Dirac delta heat source $q(\mathbf{r}) = Q\delta(\mathbf{r} - \mathbf{r}_0)$.

Assumption: Linear superposition. For an array of N nanoparticles positioned at $\mathbf{r}_i$, each absorbing power Q_i, the heat equation is fully linear. The total temperature profile is exactly the sum of the individual contributions:

$$\Delta T(\mathbf{r}) = \sum_{i=1}^{N} \frac{Q_i}{4\pi\kappa|\mathbf{r} - \mathbf{r}_i|}. \tag{2.35}$$

2.3 Green's Function for a Three-Layer Stratified Medium

In realistic experimental conditions, nanoparticles are rarely suspended in an infinite homogeneous fluid. Instead, they are typically deposited on a glass substrate and immersed in water. The presence of the substrate significantly alters the thermal diffusion landscape because the thermal conductivity of glass ($\kappa_g \approx 1.1$ W/mK) differs from water ($\kappa_w \approx 0.6$ W/mK).

Assumptions:

- Isotropic thermal conductivity κ within each layer.
- The interfaces between layers are perfectly planar.
- The layers have infinite lateral extent.
- There is no thermal contact resistance (perfect thermal boundary).

We can elegantly solve this using the method of image charges, an approach directly analogous to electrostatics Baffou et al. [2014]. Because the heat equation is a Poisson

equation outside the source, an interface between two media of differing thermal conductivities reflects and transmits "thermal flux" just like a dielectric boundary reflects electric field lines.

Consider a 3-layer system: a glass substrate (layer 1, $z < b_1$), a spacer layer containing the nanoparticle (layer 2, $b_1 \leq z \leq b_2$), and a water superstrate (layer 3, $z > b_2$). A point heat source Q is located at (x_0, y_0, z_0) in the middle layer.

We define thermal reflection and transmission coefficients analogous to Fresnel equations. For a boundary from medium i to j:

$$R_{ij} = \frac{\kappa_i - \kappa_j}{\kappa_i + \kappa_j}, \tag{2.36}$$

$$T_{ij} = \frac{2\kappa_i}{\kappa_i + \kappa_j}. \tag{2.37}$$

Let $R_{\text{bot}} = R_{21}$ be the reflection at the bottom interface and $R_{\text{top}} = R_{23}$ at the top interface. The multiple internal reflections are characterized by the loop gain factor $W = R_{\text{top}} R_{\text{bot}}$. Because thermal conductivities are strictly positive, $|R_{ij}| < 1$ for all real materials, ensuring that $|W| < 1$. This guarantees that the infinite series representations will unconditionally converge.

We define the inverse distance function to an image charge at depth ζ:

$$G_0(z, \zeta) = \frac{1}{\sqrt{\rho^2 + (z - \zeta)^2}}, \tag{2.38}$$

where $\rho = \sqrt{(x - x_0)^2 + (y - y_0)^2}$.

By tracing the infinite sequence of bouncing "image heat sources" trapped between the two parallel interfaces separated by $d = b_2 - b_1$, the temperature field in the middle layer ($b_1 \leq z \leq b_2$) is the sum of the primary source and four infinite series of reflections:

$$T_2(r) = \frac{Q}{4\pi\kappa_2} \left[G_0(z, z_0) + \sum_{n=0}^{\infty} W^n \Big(W G_0(z, \zeta_1^{(n)}) + R_{\text{top}} G_0(z, \zeta_2^{(n)}) + R_{\text{bot}} G_0(z, \zeta_3^{(n)}) + W G_0(z, \zeta_4^{(n)}) \Big) \right], \tag{2.39}$$

where the image depths are geometrically traced as:

$$\zeta_1^{(n)} = z_0 + 2(n+1)d, \tag{2.40}$$

$$\zeta_2^{(n)} = 2b_2 - z_0 + 2nd, \tag{2.41}$$

$$\zeta_3^{(n)} = 2b_1 - z_0 - 2nd, \tag{2.42}$$

$$\zeta_4^{(n)} = z_0 - 2(n+1)d. \tag{2.43}$$

The physics here is straightforward: each term represents an image heat source created by reflection off the top interface, bottom interface, or multiple bounces between them.

The amplitude of each subsequent bounce diminishes by a factor of W.

The temperature field transmitted into the bottom layer ($z < b_1$) only consists of two series (direct transmission and transmission after bouncing off the top interface):

$$T_1(r) = \frac{Q}{4\pi\kappa_2} T_{21} \sum_{n=0}^{\infty} W^n \Big(G_0(z, z_0 + 2nd) + R_{\text{top}} G_0(z, 2b_2 - z_0 + 2nd) \Big). \tag{2.44}$$

Similarly, the temperature field transmitted into the top layer ($z > b_2$) is:

$$T_3(r) = \frac{Q}{4\pi\kappa_2} T_{23} \sum_{n=0}^{\infty} W^n \Big(G_0(z, z_0 - 2nd) + R_{\text{bot}} G_0(z, 2b_1 - z_0 - 2nd) \Big). \tag{2.45}$$

This rigorous Green's tensor formulation allows us to evaluate the full three-dimensional temperature distribution of complex nanoparticle arrays accurately, accounting for all boundary effects without resorting to computationally expensive finite-element method (FEM) simulations.

2.4 The Thermo-Optic Effect and Phase Measurement

The temperature variations generated by the nanoparticles are invisible to the naked eye. To measure them, we exploit the thermo-optic effect: the refractive index of a fluid changes as a function of temperature. By passing a weak probe beam through the heated fluid and measuring the accumulated optical phase shift via interferometry (such as Quadriwave Lateral Shearing Interferometry), we can extract the temperature map Bon et al. [2009].

Assumption: The probe beam is sufficiently weak that it does not cause any photothermal heating itself; it passively senses the environment.

Assumption: The sample has a well-defined and known thickness h across which the phase is integrated.

For pure water, the relationship between refractive index and temperature is notably nonlinear, especially as one moves away from room temperature. Baffou et al. derived empirical coefficients to express this nonlinearity as a fourth-order polynomial Baffou, Rigneault, Marber, and Quidant [2020]:

$$\Delta n(T) = b_1 T + b_2 T^2 + b_3 T^3 + b_4 T^4, \tag{2.46}$$

where $b_1 = -1.0514 \times 10^{-4}$, $b_2 = -1.5692 \times 10^{-6}$, $b_3 = 5.7538 \times 10^{-9}$, and $b_4 = -1.2873 \times 10^{-11}$.

However, this raw polynomial assumes T is the absolute Celsius temperature. Because

our heat solver naturally computes the temperature *increase* $\Delta T = T - T_\infty$, it is computationally superior to re-center the Taylor expansion around the ambient temperature T_∞. We define the centered coefficients β_k:

$$\beta_1 = b_1 + 2T_\infty b_2 + 3T_\infty^2 b_3 + 4T_\infty^3 b_4, \tag{2.47}$$

$$\beta_2 = b_2 + 3T_\infty b_3 + 6T_\infty^2 b_4, \tag{2.48}$$

$$\beta_3 = b_3 + 4T_\infty b_4, \tag{2.49}$$

$$\beta_4 = b_4. \tag{2.50}$$

This allows us to evaluate the refractive index change directly from our simulated temperature map as:

$$\Delta n(\mathbf{r}) = \beta_1 \Delta T(\mathbf{r}) + \beta_2 \Delta T(\mathbf{r})^2 + \beta_3 \Delta T(\mathbf{r})^3 + \beta_4 \Delta T(\mathbf{r})^4. \tag{2.51}$$

The total optical phase shift $\Delta\phi(x, y)$ experienced by a plane wave passing through the sample in the z-direction is the path integral of the refractive index change:

$$\Delta\phi(x, y) = \frac{2\pi}{\lambda_{\text{probe}}} \int_0^h \Delta n(x, y, z)\, dz. \tag{2.52}$$

By matching this theoretical forward-model of the phase to experimental QWLSI images, we can quantitatively solve the inverse problem and pinpoint the exact nanoscale heat generation occurring in our samples.

3. FORWARD MODELING AND EXPERIMENTAL DATA PROCESSING

3.1 The FFT-Based Forward Solver

In order to train data-driven machine learning models for thermodynamic inversion, we require a rapid and highly accurate forward model. The forward model takes a given heat source density map $Q(x, y)$ and computes the resultant temperature field $T(x, y)$ and the cumulative optical phase shift $\Delta\phi(x, y)$ observed through a medium. While traditional Finite Element Method (FEM) solvers provide rigorous solutions, they are computationally prohibitive for generating large datasets encompassing tens of thousands of samples. To address this, we developed a high-performance FFT-based forward solver.

3.1.1 Spatial Convolution in Fourier Space

The steady-state heat equation for a localized source distribution $Q(x, y)$ in a homogeneous medium with effective thermal conductivity κ_{eff} can be formulated via the Green's function formalism. The temperature rise $T(x, y)$ is mathematically expressed as a spatial convolution between the heat source and the system's thermal Green's function $G_T(x, y)$:

$$T(x, y) = Q(x, y) \circledast G_T(x, y) \tag{3.1}$$

By invoking the Convolution Theorem, this spatial convolution is mapped to a simple element-wise multiplication in the spatial frequency domain (Fourier space):

$$\mathcal{F}\{T(x, y)\} = \mathcal{F}\{Q(x, y)\} \cdot \mathcal{F}\{G_T(x, y)\} \implies \hat{T} = \hat{Q} \cdot \hat{G}_T \tag{3.2}$$

where $\mathcal{F}$ denotes the 2D Fourier Transform. The temperature field is subsequently recovered via the Inverse Fourier Transform:

$$T(x, y) = \mathcal{F}^{-1}\{\hat{Q} \cdot \hat{G}_T\} \tag{3.3}$$

For a heat source at the origin on a substrate, the 3D thermal Green's function evaluated at a radial distance $R = \sqrt{x^2 + y^2}$ in the measurement plane is Baffou et al.

[2010a]:

$$G_T(R) = \frac{1}{4\pi\kappa_{\text{eff}}R} \tag{3.4}$$

Similarly, the linear component of the resulting optical phase shift integrated over a fluid layer of height h can be represented using a linear phase kernel G_ϕ:

$$G_\phi(R) = \frac{\beta_1}{4\pi\kappa_{\text{eff}}} \operatorname{arcsinh}\left(\frac{h}{R}\right) \tag{3.5}$$

where β_1 is the first-order thermo-optic coefficient ($\partial n/\partial T$).

The discrete Fourier transform (DFT) inherently assumes periodic boundary conditions, leading to circular convolution. **Assumption:** We assume periodic boundary conditions introduced by the FFT can be mitigated by zero-padding. To compute a strictly linear convolution and avoid unphysical boundary wrap-around artifacts (energy from one side of the domain wrapping to the opposite), we implement a symmetric zero-padding strategy. If our physical domain contains $N_y \times N_x$ pixels, we pad the grid symmetrically by N_y and N_x on each side, expanding the computational domain to $(3N_y) \times (3N_x)$ dynamically (or at minimum $2N_y \times 2N_x$ based on implementation, specifically padding N_y and N_x such that total size is $N_y + 2N_y = 3N_y$, or exactly adding $pad_y = N_y$ creating $N_y + 2 \cdot N_y \rightarrow 3N_y$, though code strictly implements $N2_y = N_y + 2 \cdot pad_y$ yielding $3N_y \times 3N_x$). This ensures the Green's function fully spans the extended domain and no overlapping periodic interactions occur within the Region of Interest (ROI).

3.1.2 Forward Solver Algorithm

The execution of the FFT forward solver follows a structured pipeline:
1. **Particle Assembly:** Nanoparticles are placed within the domain at defined coordinates (x_p, y_p). The required grid resolution is dynamically determined to resolve individual nanoparticles. The pixel size is calculated as $\Delta x = a/\gamma$, where a is the characteristic particle size and γ is the resolution factor (typically 3.0).
2. **Power Map Generation:** The optical power absorbed by each particle is mapped onto the grid, scaled by the local beam intensity, forming the discrete source matrix Q.
3. **Padding & Transform:** The Q matrix and the spatial kernels G_T and G_ϕ are zero-padded. Their Fast Fourier Transforms are computed.
4. **Convolution & Cropping:** Element-wise multiplication in k-space is followed by the inverse FFT. The results are symmetrically cropped back to the original $N_y \times N_x$ dimensions.

While the linear phase shift is computed directly via a 2D convolution kernel, the exact optical phase shift depends nonlinearly on the temperature profile due to higher-order terms in the thermo-optic polynomial of the fluid: $\Delta n(T) = \beta_1 T + \beta_2 T^2 + \beta_3 T^3 + \beta_4 T^4$.

To capture this rigor without full 3D simulation overhead, we evaluate the nonlinear residual. We divide the fluid thickness into discrete horizontal slices (e.g., 15 slices of thickness dz). For each slice at height z, we compute the thermal kernel $G_{T,z}$ and compute the temperature T_z via convolution. We then evaluate the higher order refractive index changes $\Delta n_{z,\text{residual}} = \beta_2 T_z^2 + \beta_3 T_z^3 + \beta_4 T_z^4$, integrating them along z:

$$\Delta\phi_{\text{residual}} = \sum_z \Delta n_{z,\text{residual}} \Delta z \tag{3.6}$$

The final phase shift is the sum of the linear component and this nonlinear residual.

3.1.3 3D Volumetric Extension

For neural network formulations utilizing Physics-Informed Neural Networks (PINNs), 3D volumetric fields are required. We extend the forward solver into 3D space natively on the GPU using PyTorch (`Native3DTQSISolver.generate_synthetic_phase`).

For an arbitrary focal plane depth z_0, the 3D Green's tensor is constructed layer-by-layer across the user-defined z-array grid (`_build_3d_greens_tensor`). The 3D temperature volume $T_{3\text{D}}$ is solved via batched FFT convolution across the z-stack.

Once the full 3D temperature distribution $T_{3\text{D}}(x, y, z)$ is obtained, we apply the location-dependent thermo-optic coefficients $\beta(z)$ corresponding to the multi-layer material stack (e.g., glass substrate, water medium, upper cover glass). The local refractive index change $\Delta n_{3\text{D}}(x, y, z)$ is evaluated using the nonlinear polynomial. Finally, the Optical Path Difference (OPD) and phase are accumulated via a cumulative sum along the axial dimension:

$$\text{OPD}_{3\text{D}}(x, y) = \sum_z \Delta n_{3\text{D}}(x, y, z) \Delta z \tag{3.7}$$

$$\Delta\phi_{3\text{D}}(x, y) = \text{OPD}_{3\text{D}}(x, y) \left(\frac{2\pi}{\lambda_{\text{probe}}} \right) \tag{3.8}$$

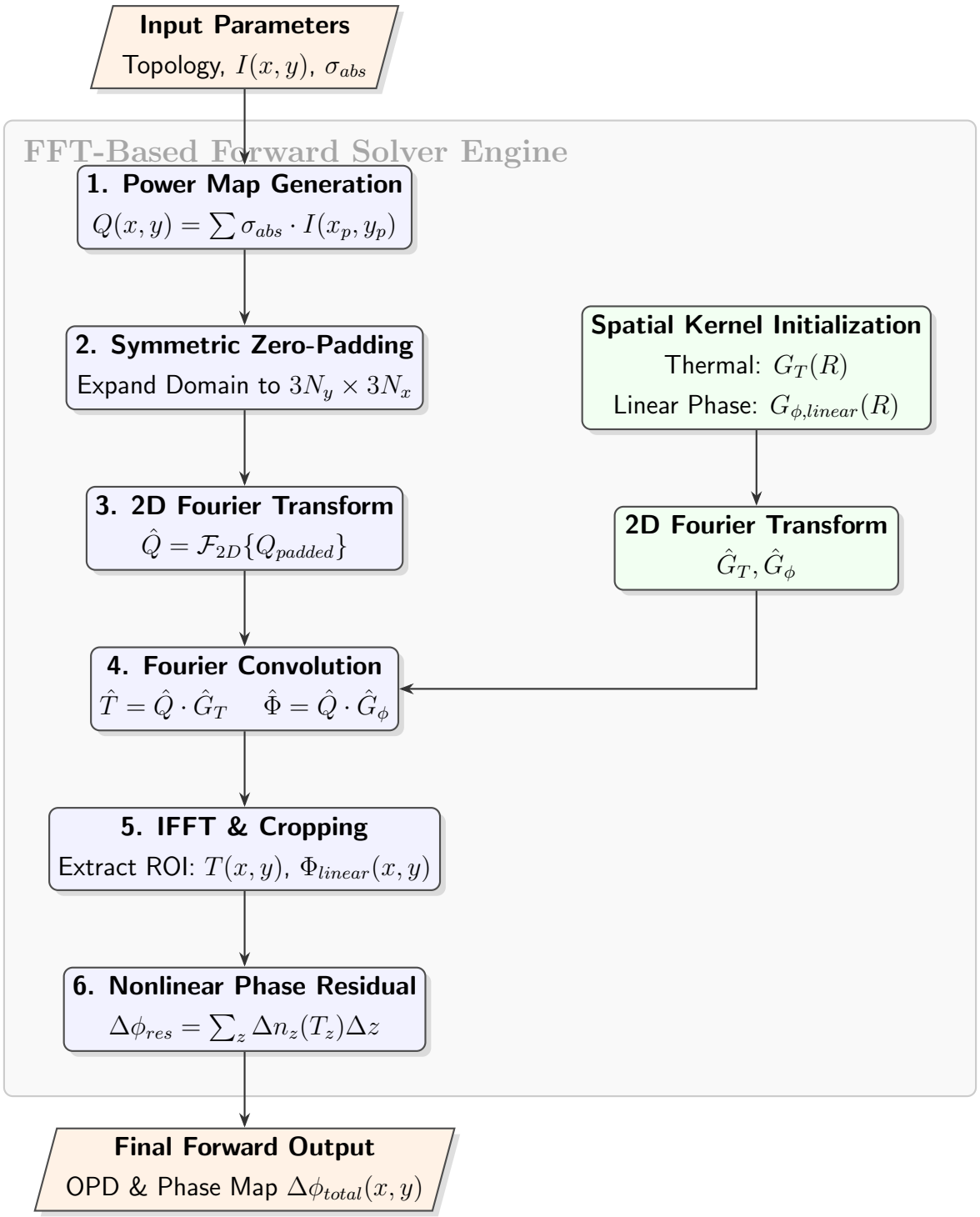


Fig. 3.1: Block diagram of the FFT-based forward solver pipeline, illustrating the generation of the power map, padding, Fourier convolution, and nonlinear phase residual computation.

3.1.4 Representative Forward Simulations for PINN Training

To generate the extensive and physically rigorous datasets required for training the Physics-Informed Neural Network (PINN) across diverse topologies, the FFT-based forward solver is utilized to compute exact spatial convolutions. The algorithm maps the

absorbed optical power into a discrete source matrix $Q(x, y)$ and convolves it with the thermal Green's function $G(x, y, z)$ evaluated for a finite-sized spherical nanoparticle in a homogeneous medium.

To avoid unphysical singularities at the origin (where $r \to 0$) and to correctly account for the particle's physical boundary, the 3D thermal Green's function utilized in the solver is strictly clamped at the particle radius a. Thus, the effective spatial kernel takes the exact form:

$$G(x, y, z) = \frac{1}{4\pi\kappa_{eff} \max\left(\sqrt{x^2 + y^2 + z^2}, a\right)} \tag{3.9}$$

For the 2D in-plane Fourier convolutions (where the heat sources and observation plane are identically at $z = 0$), this strictly evaluates to $G(x, y, 0)$. This mathematically ensures that the computed temperature rise ΔT within and exactly at the surface of the nanoparticle remains bounded to the physical maximum $\Delta T_{max} = \frac{Q}{4\pi\kappa_{eff}a}$, seamlessly transitioning into the standard $1/r$ thermal decay in the surrounding fluid.

The following figures visually demonstrate this convolution process—from the isolated point-source inputs and spatial kernels to the final steady-state temperature fields $\Delta T(r)$. These high-fidelity synthetic maps form the foundational ground-truth data used to train the neural network's spatial extrusion and thermodynamic regularization modules.

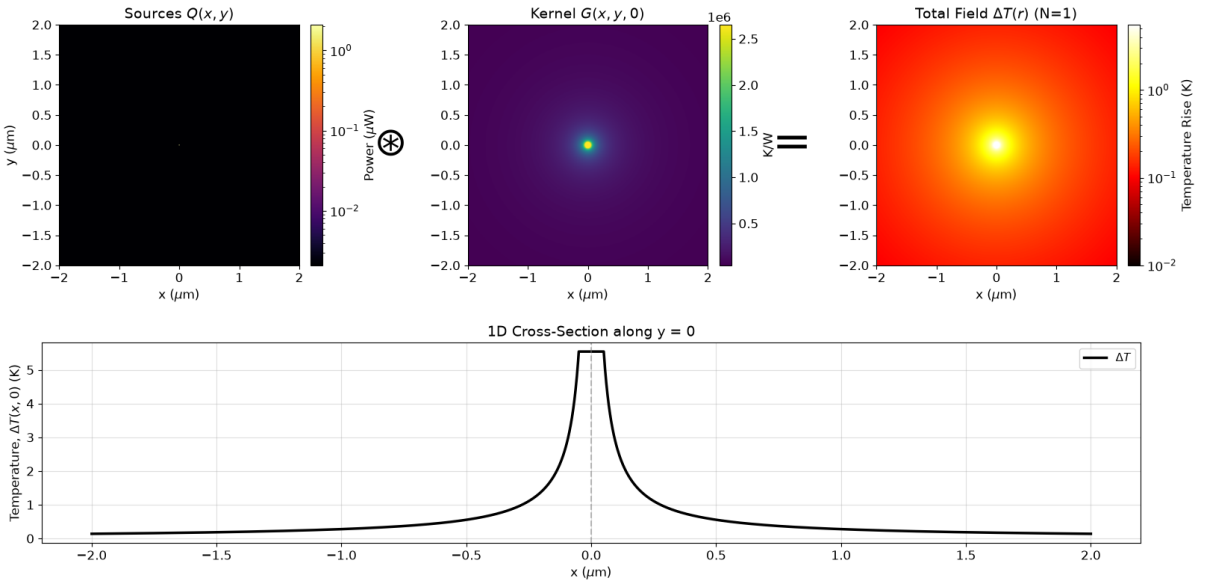


Fig. 3.2: Forward solver simulation for a **Single Isolated Nanoparticle**. *Specifications:* Gold (Au) nanosphere, radius $a = 50$ nm; Ambient medium: Homogeneous Water ($\kappa_{eff} = 0.60$ W/m·K, $n_{med} = 1.33$); Particle refractive index $n_s = 0.54 + 2.23j$; Illumination: Uniform plane wave at $\lambda = 532$ nm, Intensity $I_0 = 100.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 2.0894 \times 10^{-2}\mu$m^2. The middle panel visualizes the finite-sized thermal Green's function kernel $G(x, y, 0)$ bounded at $r = a$, preventing a central singularity. The bottom panel plots the strict $1/r$ thermal decay profile dropping smoothly from the surface peak ΔT along the 1D cross-section at $y = 0$.

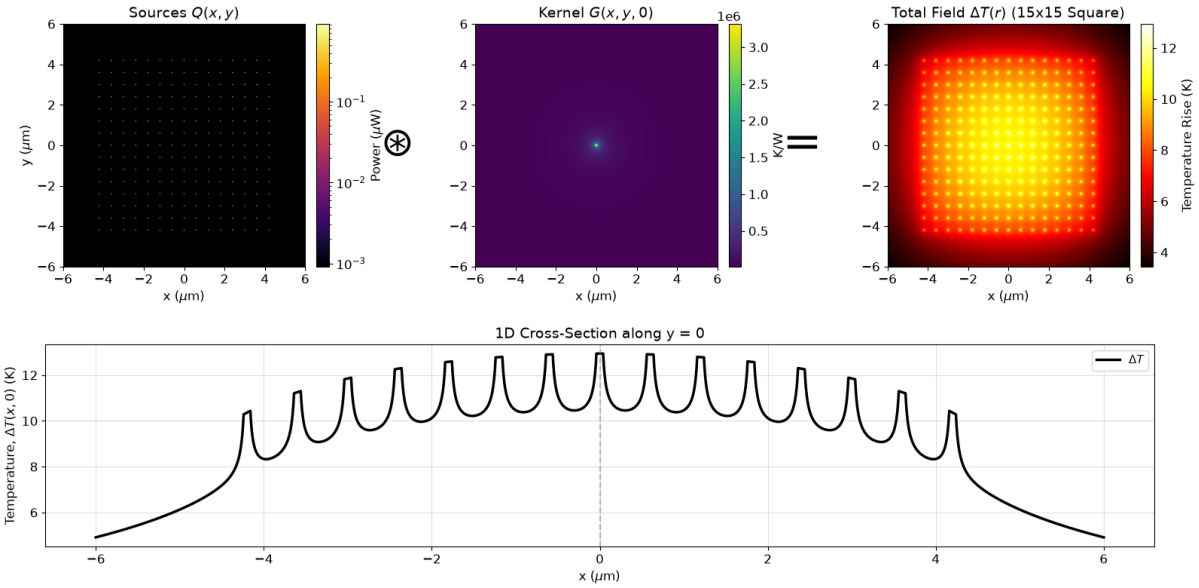


Fig. 3.3: Forward solver simulation for a **Square Nanoparticle Array under Uniform Illumination**. *Specifications:* Gold (Au) nanospheres, radius $a = 40$ nm; Array dimensions: 15×15 particles with lattice pitch $p = 600$ nm; Ambient medium: Homogeneous Water ($\kappa_{eff} = 0.60$ W/m·K, $n_{med} = 1.33$); Particle refractive index $n_s = 0.54 + 2.23j$; Illumination: Uniform plane wave at $\lambda = 532$ nm, Constant Intensity $I_0 = 50.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 1.8289 \times 10^{-2} \mu$m^2. The discrete source matrix $Q(x, y)$ mirrors the flat-top spatial envelope, resulting in a broad collective heating profile superimposed with uniform, highly localized intra-particle thermal spikes approaching a ΔT of 13 K.

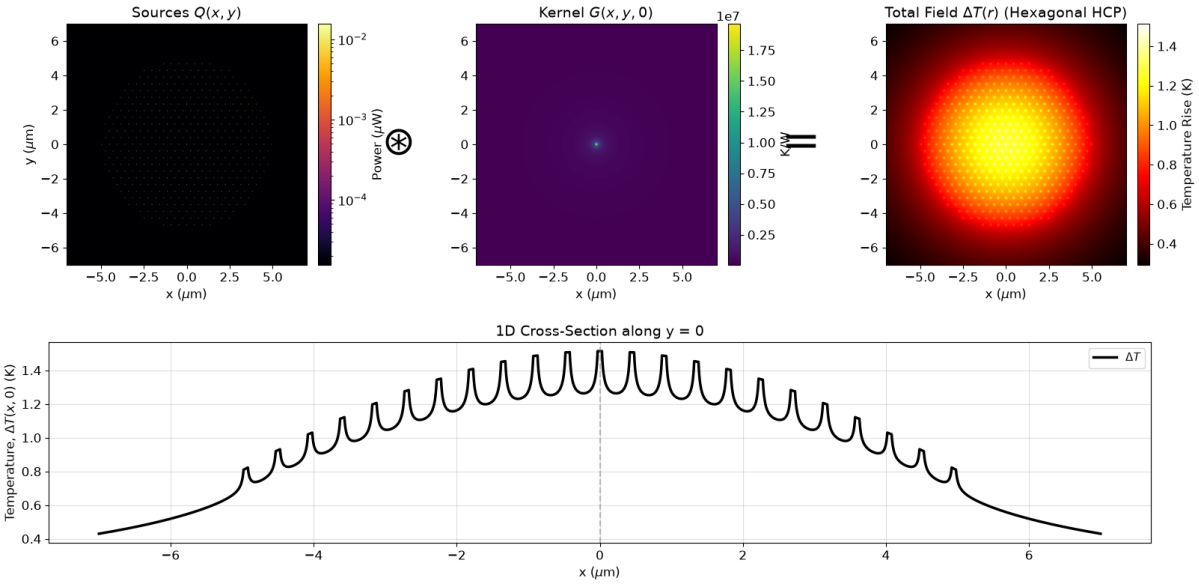


Fig. 3.4: Forward solver simulation for a **Hexagonal Lattice under Gaussian Illumination**. *Specifications:* Silver (Ag) nanospheres, radius $a = 30$ nm; Hexagonal close-packed (HCP) 2D lattice bounded by a 10 μm circular aperture with nearest-neighbor pitch $p = 450$ nm; Ambient medium: Isopropanol (IPA, $\kappa_{eff} = 0.135$ W/m·K, $n_{med} = 1.37$); Particle refractive index $n_s = 0.05 + 3.42j$; Illumination: Gaussian beam profile at $\lambda = 532$ nm, beam waist $w_0 = 8.0\mu$m, Peak Intensity $I_0 = 50.0$ mW/μm^2; Exact Mie theory absorption cross-section $\sigma_{abs} = 3.1174 \times 10^{-4} \mu$m^2. The lower thermal conductivity of the IPA medium strongly restricts heat dissipation compared to water, heavily amplifying the macroscopic collective thermal envelope across the dense lattice.

3.2 Experimental Data Processing Pipeline

To validate our computational models and train robust machine learning algorithms, we process experimental data acquired via Off-Axis Digital Holographic Interferometry (DHI) Baffou et al. [2012], Bon et al. [2009].

3.2.1 Off-Axis Digital Holographic Interferometry

In off-axis holography, an angle is introduced between the probe and reference beams. This causes the interferogram to exhibit distinct spatial frequencies. The experimental raw data from the camera is recorded in high-throughput TDMS (Technical Data Management Streaming) files. The initial processing reshapes the flattened 1D raw data streams into interleaved image stacks matching the sensor geometry.

In the spatial frequency domain of the hologram, three distinct regions emerge: the zero-order DC term representing the background intensity, and two conjugate first-order sidebands which carry the complex optical field of the sample. The off-axis geometry displaces these sidebands away from the DC core, enabling their isolation.

3.2.2 Sideband Isolation via Super-Gaussian Windowing

Extracting the complex field requires masking out the DC component and isolating one of the sidebands. We utilize a Super-Gaussian windowing function $W(r)$ applied in Fourier space:

$$W(r) = \exp\left(-\left(\frac{r}{r_0}\right)^4\right) \tag{3.10}$$

where r_0 defines the window radius. The fourth-order exponent provides a spatial filter that is sharper than a standard Gaussian (preventing excessive attenuation of high spatial frequencies near the sideband edge) yet smoother than a strict top-hat filter (preventing ringing artifacts known as Gibbs phenomenon during the inverse transform).

The algorithm automatically masks the DC component and identifies the peak magnitude in the right half-plane. The Super-Gaussian window is centered at this peak. Inverse Fourier Transformation (IFFT) of this isolated window yields the complex optical field E.

To compute the pump-induced thermodynamic phase shift, we calculate the phase difference between states when the pump laser is 'ON' and 'OFF'. To maximize the signal-to-noise ratio (SNR) across thousands of frames, we accumulate the phase shift via the complex conjugate frame-pair product:

$$E_{\text{avg}} = \sum_{i=1}^{N_{\text{pairs}}} E_{\text{on}}^{(i)} \times \left(E_{\text{off}}^{(i)}\right)^* \tag{3.11}$$

The angle of this accumulated complex field provides the raw phase shift. **Assumption:** Heating reduces the refractive index of water ($\partial n/\partial T < 0$), which yields a negative phase shift. We flip the sign for convention so that the extracted phase intuitively represents a positive temperature rise.

3.2.3 Automated Camera Harmonic Noise Detection

Experimental measurements are often plagued by structural periodic noise originating from camera sensor harmonics, which manifest as anomalous high-intensity speckles in the Fourier plane. We developed a fully automated, robust algorithmic pipeline to detect and eliminate this harmonic noise without requiring manual user intervention.

The pipeline comprises the following sequential steps:

1. **Azimuthal Profile Computation:** The radial background profile $B(r)$ is computed by azimuthally averaging the logarithmic magnitude of the Fourier space, $\log|F(k)|$.
2. **Residual Map Calculation:** The spatially-varying background is subtracted to form a residual image: $R(x, y) = \log|F(x, y)| - B(r(x, y))$.
3. **Exclusion Zones:** We define protected regions where noise detection is strictly

disabled to prevent stripping out legitimate signal. These include a radius around the DC core, a radius around the isolated sideband, and the vertical/horizontal cross-lines emanating from the DC center.

4. **Robust Sigma Estimation (MAD):** Within discrete annular rings, we estimate the local noise variance. Because anomalous spots significantly skew standard deviation metrics, we use the Median Absolute Deviation (MAD):

$$\sigma_{\text{robust}} = 1.4826 \times \text{median}(|R_i - \text{median}(R_i)|) \tag{3.12}$$

5. **Adaptive Thresholding:** A pixel is flagged as a candidate anomaly if its residual significantly exceeds the local noise floor: $R(x, y) > \sigma_{\text{threshold}} \times \sigma_{\text{robust}}(r)$.

6. **Non-Maximum Suppression (NMS):** To isolate discrete spots, a 5×5 maximum filter suppresses all but the local maxima.

7. **DBSCAN Clustering:** Multi-pixel detections belonging to the same noise feature are geometrically clustered using the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. The brightest pixel in each cluster is identified as the spot center.

8. **Notch Filtering:** A circular mask (notch filter) is applied to zero out the identified spots in the complex Fourier space prior to IFFT reconstruction. The automated cleaning pipeline can be executed and tuned programmatically. We can control the `sigma_threshold` parameter to specify the sensitivity of the noise detection. A low sigma implies high sensitivity, leading to more detected spots, while a high sigma implies low sensitivity, resulting in fewer spots. If the `noise_specs` parameter is provided (i.e., $\neq$ `None`), it automatically triggers manual cleaning mode, where explicit coordinates can be passed. Additionally, the radius of the masking spots (`spot_radius_um`) can be adjusted to preserve the legitimate sample response.

```
print("Running Automated Pipeline Diagnostics...")
phase_auto = NanoThermalSolver.process_and_clean_tdms(
    tdms_path,
    noise_specs=None,                # Triggers Automated Mode
    show_pipeline=True,              # <-- THIS GENERATES THE 12-
       PANEL PLOT
    sigma_threshold=3.0,             # controls the sensitivity of
       the outlier detection
    dc_exclusion_radius_px=2,
    sb_exclusion_radius_px=45,
    probe_lambda_nm=457.0,           # Automatically returns true
       OPD (nm)
    spot_radius_um= 4                # controls the size of the
```

```
        notch filter applied to each detected spot
)
```

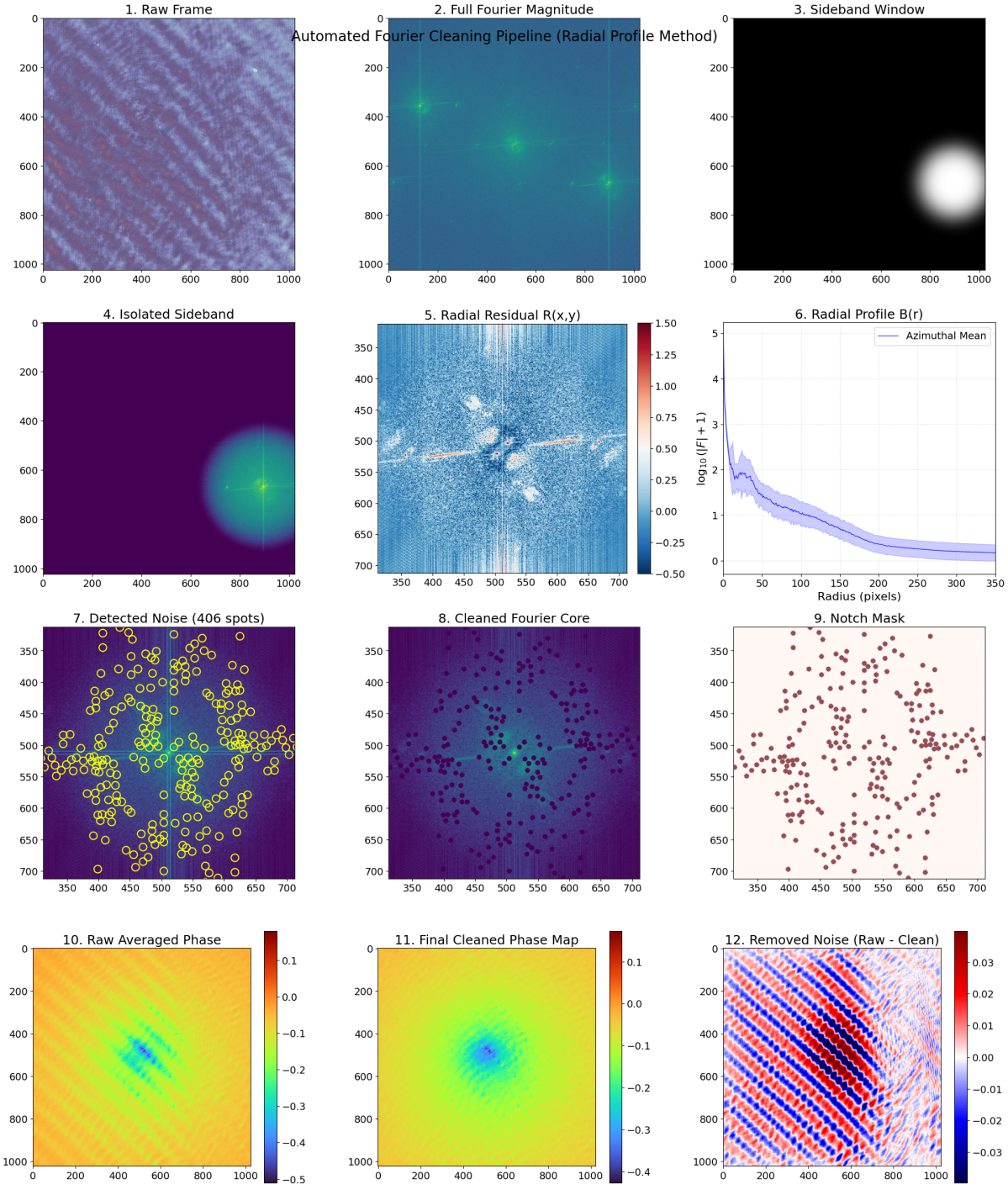


Fig. 3.5: Detailed steps of the TDMS data cleaning procedure. The 12-panel plot illustrates the full diagnostic pipeline, detecting harmonic noise and filtering it out in the Fourier domain.

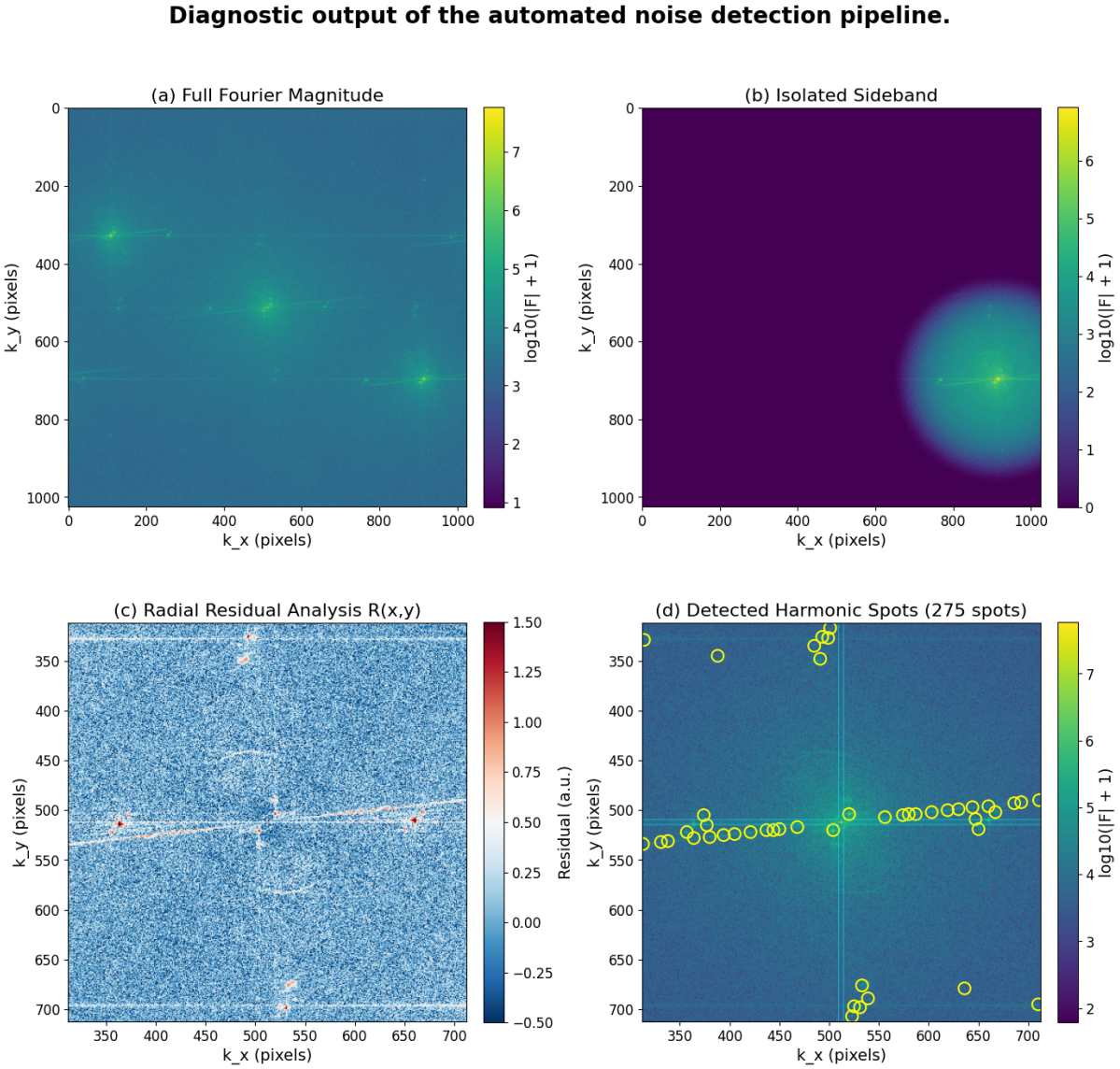


Fig. 3.6: Comparison of the phase map before and after the automated cleaning procedure, showcasing the removal of periodic artifacts while preserving the true sample response.

3.2.4 Phase Extraction and OPD Conversion

After the spatial frequencies have been cleanly reconstructed, the resulting phase shift array $\Delta\phi$ (in radians) is converted into the physical Optical Path Difference (OPD, in nanometers) using the known wavelength of the probe laser λ_{probe}:

$$\text{OPD}_{\text{nm}} = \Delta\phi_{\text{rad}} \times \left(\frac{\lambda_{\text{probe}}}{2\pi} \right) \tag{3.13}$$

3.3 Synthetic Dataset Generation for Neural Network Training

To train data-driven models generalizing across varied thermodynamic architectures, we developed an extensive dataset generation pipeline leveraging our fast forward solver. The pipeline introduces severe variations in topology, material boundaries, and illumination conditions.

3.3.1 Ultra-Diverse Topology Engine

The topology engine synthesizes nanoparticle layouts spanning four fundamental geometric classes:

- **Bravais Lattices:** Highly symmetric arrays (square, hexagonal, rectangular, and oblique) representing engineered metamaterials.
- **Standard Shapes:** Geometric formations including continuous rings, linear chains, crosses, and triangles.
- **Amorphous Random:** Completely stochastic particle placement, analogous to colloidal suspensions or unpatterned rough films.
- **Multi-Cluster:** Several dense, highly localized islands of nanoparticles separated by sparse voids.

To induce spatial chaos and ensure the model does not overfit to perfect symmetry, these layouts are subjected to structural jitter (random local displacements of particles), global rotational transformations, and spatial shifting.

[PLACEHOLDER FIGURE]

Fig. 3.7: Gallery of generated synthetic topologies showcasing Bravais lattice, geometric ring, amorphous random distribution, and multi-cluster arrays.

3.3.2 Physical Parameter Randomization

The synthetic datasets must reflect the diverse material parameter space encountered experimentally. A robust randomization protocol is applied during batch generation.

The dielectric background is assembled as a multi-layer stack. Materials are randomly drawn from a defined chemical database encompassing glass, silica, water, ethanol, isopropyl alcohol (IPA), PDMS, and acetone. The exact axial positions of the layer boundaries (b_1 and b_2) are stochastically varied. Nanoparticles are initialized with five possible

z-placement strategies: exactly on the b_1 interface, on the b_2 interface, or suspended freely within any of the three distinct fluid/substrate layers.

Illumination is simulated utilizing a spatially variable profile engine. While simple uniform plane waves are generated, we also incorporate Gaussian beam profiles and Laguerre-Gaussian (donut) modes, significantly intensifying the spatial variance of the resultant heat source matrix $Q(x, y)$. **Assumption:** Nanoparticles are modeled as point heat sources. This is a valid approximation because the nanoparticle radius R is much smaller than the typical inter-particle spacing. Nanoparticles are modeled as either Gold or Silver spheres with radii varying continuously between 20 nm and 100 nm. The effective absorption cross-section (σ_{abs}) is computed rigorously using Mie theory for the selected pump wavelength (532 nm or 808 nm) relative to the local refractive index.

[PLACEHOLDER FIGURE]

Fig. 3.8: Placeholder for a table listing the materials database, detailing respective thermal conductivities (κ), refractive indices (n), and thermo-optic polynomials (β).

3.3.3 Adaptive Temperature Rescaling

Plasmonic arrays can reach varying absolute temperatures depending on experimental pump power. To ensure the neural network is exposed to a uniformly distributed dynamic range, an adaptive rescaling mechanism is implemented post-simulation.

Once the initial 3D temperature volume T_{3D} is solved for a normalized intensity, the peak temperature is evaluated. We dynamically draw a random target maximum temperature uniformly distributed in the interval $[5\ \text{K}, 2000\ \text{K}]$. A linear scaling factor is derived and applied identically to both T_{3D} and the heat source matrix Q.

Because the optical phase shift relies on the non-linear thermo-optic polynomial, it cannot simply be linearly scaled. Thus, following the uniform scaling of the temperature volume, the full nonlinear phase shift must be explicitly recomputed via integration across the fluid domains using the $\beta(z)$ properties of the random multi-layer stack. Finally, a strict validation step confirms numeric stability, automatically rejecting samples exhibiting non-finite fields or near-zero signal amplitudes.

4. PHYSICS-INFORMED DEEP LEARNING FOR THE INVERSE PROBLEM

4.1 The Failure of Standard Analytical Inversion

The fundamental challenge in quantitative phase imaging of thermal fields lies in the fact that the optical phase difference (OPD) is an integrated measurement along the axial direction (the z-axis). When a light beam passes through a non-uniform temperature distribution $T(x, y, z)$, the accumulated phase is given by the integral of the refractive index change. In the inverse problem, our goal is to recover the 3D temperature field $T(x, y, z)$ or the corresponding volumetric power density $p(x, y, z)$ from a single 2D phase map $\Delta\phi(x, y)$.

4.1.1 Linear Tikhonov Deconvolution

In the small temperature limit, the relationship between the phase map and the heat source distribution can be approximated as a linear convolution. The continuous space representation is often discretised and solved in the spatial frequency domain using the convolution theorem. Let the linear optical phase Green's function be denoted by $\hat{G}_l(k_x, k_y)$ and the measured phase by $\hat{P}(k_x, k_y)$ in the Fourier domain. The inverse problem is notoriously ill-posed due to zeros or small values in the transfer function, requiring regularization. Using standard Tikhonov regularization, the power density can be estimated as:

$$\hat{P}_{\text{est}}(k_x, k_y) = \frac{\hat{\Phi}(k_x, k_y) \cdot \hat{G}_l^*(k_x, k_y)}{|\hat{G}_l(k_x, k_y)|^2 + \alpha} \tag{4.1}$$

where α is the regularization parameter, $\hat{\Phi}$ is the Fourier transform of the measured phase, and $*$ denotes the complex conjugate. This approach efficiently recovers the 2D power density for thin planar sources Baffou et al. [2012, 2014].

However, this linear inversion fundamentally fails for complex 3D structures. The integration of the optical phase completely collapses the z-dimensional information into a single OPD value. Consequently, a hot heat source located shallowly within the medium

can produce an identical phase signature to a cooler, broadly distributed source located deeper in the medium. The linear deconvolution method lacks the necessary depth-resolving capabilities to decouple this ambiguity.

4.1.2 Iterative Nonlinear Inversion

For larger temperature rises ($\Delta T \gtrsim 20$ K), the thermo-optic coefficient $\partial n/\partial T$ is no longer constant, and the heat equation is coupled with a non-linear phase accumulation. To address this, iterative non-linear inversion schemes are employed. The method iteratively updates the estimated power density $p^{(i)}$ using a gradient-descent-like step scaled by a damping factor g:

$$p^{(i+1)} = p^{(i)} + g \cdot \mathcal{F}^{-1}\left(\frac{\mathcal{F}(\Delta\phi^{(i)}_{\text{error}}) \cdot \hat{G}^{*}_{l}}{|\hat{G}_l|^2 + \alpha}\right) \tag{4.2}$$

where $\Delta\phi^{(i)}_{\text{error}} = \Delta\phi_{\text{measured}} - \Delta\phi_{\text{forward}}(p^{(i)})$ is the residual phase error at the i-th iteration.

While this iterative scheme accurately accounts for the non-linear thermo-optic effect, it suffers from the same geometric limitation as the linear solver: it is strictly a 2D inversion. It assumes a known, fixed z-plane for the heat sources, making it impossible to distinguish between varying depth profiles or volumetric distributions.

4.1.3 Why a Neural Network?

The inherent z-ambiguity makes the inverse problem heavily underdetermined; multiple 3D temperature fields can yield the exact same 2D phase map. Purely analytical deconvolution methods cannot break this degeneracy. We must instead inject prior knowledge about the physical structure of typical temperature fields into the solver.

We need to be honest here: the neural network is a learned prior, not a physics solver. By training on a diverse dataset of simulated 3D thermal fields and their corresponding 2D phase maps, the network learns spatial correlations and the distinct signatures of heat diffusion from specific 3D geometries. The physics loss prevents the network from outputting physically impossible fields, but it does not replace solving the PDE. **Assumption:** We assume steady-state conditions, meaning transient thermal dynamics are ignored and the steady-state heat equation $\nabla \cdot (\kappa \nabla T) = 0$ applies. By incorporating this physics-informed loss function during training, we restrict the manifold of possible solutions to physically valid fields, effectively breaking the ambiguity Raissi et al. [2019], Karniadakis et al. [2021].

4.2 Network Architecture and Pipeline: The Physics-Guided Inversion UNet

Before delving into the specific sub-modules of the network, it is essential to understand the complete end-to-end computational pipeline. The process transitions from generating synthetic physical priors to performing deep-learning-based inversions.

4.2.1 Pipeline Overview and Constraints

The pipeline is fundamentally constrained by the knowns and unknowns of the system.

- **Knowns:** The 2D integrated optical phase map $\Delta\phi(x, y)$, the system's geometry, the 3D thermal conductivity tensor $\kappa(x, y, z)$, and the 3D thermo-optic coefficient tensor $\beta(x, y, z)$.
- **Unknowns:** The true 3D temperature field $T(x, y, z)$, the volumetric power density $p(x, y, z)$, and crucially, the exact z-location of the heat sources.
- **Constraints:** The temperature field must strictly obey the steady-state heat equation $\nabla \cdot (\kappa \nabla T) = 0$ outside the source regions. Furthermore, $\Delta T \geq 0$ everywhere, and the phase map must perfectly match the z-integrated change in refractive index.

In our workflow, the `nanothermal_3D_lib.py` script acts as the forward modeling engine. It takes physical configurations (like multi-particle arrays in a single plane) and exactly solves the heat equation using Fourier-domain Green's functions to generate a 3D phase tensor and extract the corresponding 2D phase map. The `PGI_Hnet.py` script then handles the inverse retrieval.

[PLACEHOLDER FOR FIGURE 4.1]

Fig. 4.1: Detailed illustration of the complete end-to-end pipeline. Multiple heat sources residing in the source plane generate a 3D thermal field. The optical accumulation yields the 2D phase map $\Delta\phi(x, y)$. The architecture processes this alongside the strictly structured 3D physical tensors (κ and β_{3D}) using spatial extrusion to fuse dimensions, subsequently predicting the spatial shape map and dynamic magnitude independently to resolve the z-location ambiguity.

4.2.2 Code Usage Example

```
[PLACEHOLDER FOR CODE USAGE EXAMPLE]
```

Listing 4.1: Example of invoking the PINN architecture for 3D temperature retrieval.

[PLACEHOLDER FOR FIGURE 4.2]

Fig. 4.2: Neuronic map of the Physics-Guided Inversion UNet, explicitly detailing the feature connections, skip layers, and the crucial bifurcation into the magnitude scaler and the spatial shape sigmoid map.

4.2.3 Design Philosophy

[PLACEHOLDER FOR DESIGN PHILOSOPHY]

4.2.4 The 2D Phase Encoder

The 2D phase encoder processes the normalized experimental optical phase map. The input is first max-normalized to preserve relative morphological information while discarding absolute scale dependencies:

$$\Delta\phi_{\text{norm}} = \frac{\Delta\phi}{\max|\Delta\phi| + \epsilon} \tag{4.3}$$

The encoder consists of two `Conv2d` layers, each followed by `GroupNorm` (4 groups) and a `SiLU` activation function. This fully convolutional block outputs a feature map with $2 \times b_f$ channels (where b_f is the base feature count) at the full spatial resolution of the input.

4.2.5 The 3D Physical Prior Encoder

The 3D encoder processes three critical volumetric priors:

1. A normalized approximate 3D temperature field $T_{\text{inv}}/T_{\text{inv,max}}$ (often a simple Gaussian extrusion of the 2D linear estimate).
2. The 3D thermal conductivity map $\kappa(x, y, z)$.
3. The 3D thermo-optic coefficient map $\beta(x, y, z)$.

Notably, typical thermo-optic coefficients for water and glass are on the order of 10^{-4} K^{-1}. Feeding such small values directly into the network causes numerical underflow and poor weight initialization dynamics. Thus, β is scaled by a factor of 10^4 to bring its variance to $\mathcal{O}(1)$. A single `Conv3d` block with `GroupNorm` and `SiLU` transforms this 3-channel input into a $2 \times b_f$ channel 3D feature volume.

4.2.6 SpatialExtrusionFusion: Bridging 2D to 3D

The critical operation bridging the 2D observation with the 3D domain is the `SpatialExtrusionFusion` module. The 2D features $f_{\text{2D}}(x, y)$ must interact with the 3D features $f_{\text{3D}}(x, y, z)$. To

achieve this, the 2D feature map is unsqueezed along the z-dimension and expanded to match the depth Z of the 3D volume:

$$f_{\text{2D, extruded}}(x, y, z) = f_{\text{2D}}(x, y) \quad \forall z \in [1, Z] \tag{4.4}$$

This extruded tensor is concatenated channel-wise with the 3D prior features. A $1 \times 1 \times 1$ 3D convolution (`Conv3d`) fuses these representations, creating a volumetric feature space where the 2D phase information acts as a conditioning signal at every depth slice.

4.2.7 The 3D U-Net Decoder

The fused volumetric representation is passed through a 3D U-Net Ronneberger, Fischer, and Brox [2015]. The encoder path uses `MaxPool3d` for downsampling and `ResBlock3D` layers (comprising dual $3 \times 3 \times 3$ convolutions with strict zero-padding to prevent boundary ringing, `GroupNorm`, and `SiLU` activations).

The bottleneck splits into two distinct processing paths. The spatial decoder path uses nearest-neighbor interpolation for upsampling, concatenates skip connections from the encoder to preserve high-frequency spatial details, and refines the features using sequential `ResBlock3D` units. The final layer is a $1 \times 1 \times 1$ convolution mapped through a strict `Sigmoid` activation, ensuring the output shape map $S(x, y, z)$ is strictly bounded within $[0, 1]$.

4.2.8 The Magnitude-Aware Output

Decoupling the prediction of the spatial shape from the absolute temperature scale stabilizes training. Why does this decoupling help? The sigmoid activation bounds the spatial shape map to $[0, 1]$, preventing gradient explosions during optimization. Meanwhile, the separate magnitude branch handles the massive dynamic range of peak temperatures we encounter (which can span from 5 K up to 2000 K).

The magnitude branch pools the bottleneck features globally using `AdaptiveAvgPool3d` to extract a holistic representation of the domain. This 1D vector is processed by a multilayer perceptron (MLP) with a `Softplus` final activation, predicting a non-negative scale adjustment factor γ_{scale}.

The dynamic peak temperature is computed relative to the maximum of the input prior $T_{\text{inv,max}}$:

$$T_{\text{peak}} = T_{\text{inv,max}} \times (\gamma_{\text{scale}} + 1.0) \tag{4.5}$$

The final absolute temperature field is then recovered as:

$$T_{\text{pred}}(x, y, z) = S(x, y, z) \times T_{\text{peak}} \tag{4.6}$$

This architecture forces the deep U-Net to focus exclusively on morphological shaping, while the magnitude branch captures the large-scale energetic variations.

4.3 The FastAdaptiveLagrangianLoss: Physics as Regularization

[PLACEHOLDER FIGURE]

Fig. 4.3: Decomposition of the FastAdaptiveLagrangianLoss into purely data-driven, structural, and physics-based penalty terms.

The success of Physics-Informed Neural Networks (PINNs) depends entirely on the design of the loss function. Our `FastAdaptiveLagrangianLoss` dynamically balances data fidelity with stringent physical constraints across the 3D domain.

4.3.1 The 3D Finite-Difference Laplacian Kernel

To enforce the steady-state heat equation, we must compute the Laplacian $\nabla^2 T$. Using automatic differentiation for 3D volumetric fields is highly memory intensive. Instead, we implement a differentiable 3D finite-difference stencil applied via standard 3D convolution (`F.conv3d`).

Because our computational grid often has anisotropic voxel dimensions ($\Delta x = \Delta y \neq \Delta z$), we define the grid aspect ratio $\gamma = \Delta x / \Delta z$. The customized $3 \times 3 \times 3$ Laplacian kernel W is defined such that the central weight is $-2(2+\gamma^2)$. The immediate neighbors in the xy-plane have a weight of $+1$, and the immediate neighbors along the z-axis have a weight of $+\gamma^2$. Explicitly, the non-zero elements of the kernel tensor are:

$$W(0,0,0) = -2(2+\gamma^2) \tag{4.7}$$

$$W(\pm 1,0,0) = 1 \quad \text{(x-neighbors)} \tag{4.8}$$

$$W(0,\pm 1,0) = 1 \quad \text{(y-neighbors)} \tag{4.9}$$

$$W(0,0,\pm 1) = \gamma^2 \quad \text{(z-neighbors)} \tag{4.10}$$

This kernel provides a highly optimized, fully back-propagatable measurement of the local curvature of the temperature field.

4.3.2 Loss Decomposition

The total loss is a weighted sum of several decoupled objective functions, all operating on the max-normalized temperature fields to maintain stable gradients regardless of the absolute temperature scale.

1. Source-Weighted L1 Loss ($w = 5.0$): Focuses the network's attention on correctly resolving the high-temperature regions (the active heat sources). L1 is chosen over L2 (MSE) to prevent blurring of sharp features.

$$\mathcal{L}_{\text{src}} = 5.0 \times \frac{\sum |T_{\text{pred}} - T_{\text{true}}| \odot M_{\text{src}}}{\sum M_{\text{src}}} \tag{4.11}$$

where M_{src} is a binary mask indicating source regions.

2. Background L1 Loss ($w = 2.0$): Ensures that the far-field regions correctly decay to ambient temperature.

$$\mathcal{L}_{\text{bg}} = 2.0 \times \frac{\sum |T_{\text{pred}} - T_{\text{true}}| \odot (1 - M_{\text{src}})}{\sum (1 - M_{\text{src}})} \tag{4.12}$$

3. Boundary Constraint Loss ($w = 2.0$): Enforces proper Dirichlet and Neumann boundary conditions by strictly penalizing deviations on the 6 outer faces of the 3D volume.

$$\mathcal{L}_{\text{bc}} = 2.0 \times \sum_{f \in \text{faces}} \text{L1}(T^{(f)}_{\text{pred}}, T^{(f)}_{\text{true}}) \tag{4.13}$$

4. Gradient Matching Loss ($w = 2.0$): A form of Total Variation matching. By penalizing the first derivatives, this term ensures the predicted field preserves sharp thermal gradients and accurately mirrors the heat flux vector field $\mathbf{q} = -\kappa \nabla T$.

$$\mathcal{L}_{\text{grad}} = 2.0 \times (\text{L1}(\partial_x T_{\text{pred}}, \partial_x T_{\text{true}}) + \text{L1}(\partial_y T_{\text{pred}}, \partial_y T_{\text{true}})) \tag{4.14}$$

5. Z-Localization Loss ($w = 2.0$): To break the depth degeneracy, the network must accurately identify the z-plane containing the peak temperature. We apply an `AdaptiveMaxPool2d` over local 5×5 regions to extract the peak amplitude profiles along the z-axis, forcing the network to match the true depth profile.

$$\mathcal{L}_{\text{zloc}} = 2.0 \times \text{L1}(\text{Peaks}_z(T_{\text{pred}}), \text{Peaks}_z(T_{\text{true}})) \tag{4.15}$$

6. Laplacian PDE Loss ($w = 0.05$): This is the fundamental physics constraint. **Assumption:** We enforce $\nabla^2 T = 0$ only in source-free regions. The source plane is explicitly excluded from the Laplacian penalty. Outside of the active heat sources, the steady-state heat equation dictates that the divergence of the heat flux must be zero: $\nabla \cdot (\kappa \nabla T) = 0$. Using our custom Laplacian stencil:

$$\mathcal{L}_{\text{lap}} = \frac{\sum |\kappa \cdot \nabla^2 T_{\text{pred}}| \odot (1 - M_{\text{src}})}{\sum (1 - M_{\text{src}})} \tag{4.16}$$

The total loss is strictly given by:

$$\mathcal{L} = \mathcal{L}_{\text{bg}} + \mathcal{L}_{\text{src}} + \mathcal{L}_{\text{bc}} + \mathcal{L}_{\text{grad}} + \mathcal{L}_{\text{zloc}} + 0.05 \times \mathcal{L}_{\text{lap}} \tag{4.17}$$

4.3.3 Why the Laplacian Weight is 0.05

The weighting of the physics loss is highly non-trivial. If the weight is too high (e.g., $w \approx 1.0$), the network becomes over-constrained by the PDE and fails to learn the complex morphological details dictated by the data terms, often collapsing to smooth, featureless harmonic functions. If the weight is too low, the physical regularization provides no meaningful guidance. Empirical tuning revealed that a weight of 0.05 provides the optimal balance, acting as a soft regularization that pulls the solution toward physical compliance without overriding the high-frequency structural data.

4.4 The Z-Guided ThermalUNet: An Alternative Architecture

An alternative architectural paradigm explored was the *Z-Guided ThermalUNet*. Rather than implicitly learning the depth via spatial extrusion and localized peak matching, this architecture explicitly predicts the z-plane of the heat source. A dedicated classification head extracts features from the 2D encoder and computes a probability distribution across all possible z-slices via a `Softmax` function.

These probabilities are used to dynamically weight the extrusion of the 2D features into the 3D volume, effectively telling the 3D decoder exactly where to place the heat source. Accordingly, the loss function replaces the L1 peak matching with a categorical Cross-Entropy loss for z-localization:

$$\mathcal{L}_{\text{z_ce}} = \text{CrossEntropy}(\mathbf{z}_{\text{logits}}, z_{\text{true_idx}}) \tag{4.18}$$

While explicit and robust, this approach assumes a single primary heat source plane, whereas the *PhysicsGuidedInversionUNet* generalizes better to volumetric heat distributions.

4.5 Training Methodology

[PLACEHOLDER FIGURE]

Fig. 4.4: Training metrics over 100 epochs, demonstrating rapid convergence of the data loss and stabilization of the physics Laplacian loss.

4.5.1 Mixed-Precision Training with Gradient Accumulation

Training 3D convolutional networks demands massive computational resources. To operate within a stringent 4GB VRAM constraint, we employ Automatic Mixed Precision (AMP) using `torch.amp.autocast`, scaling gradients via a `GradScaler`. Mixed precision casts computationally heavy operations (like 3D convolutions) into `float16` while keeping critical accumulations in `float32`.

Due to memory limits, the physical batch size is restricted to exactly 1. To maintain a stable estimate of the gradient, we utilize gradient accumulation over 4 steps, yielding an effective batch size of 4. Furthermore, to prevent catastrophic gradient explosions driven by the complex PDE loss landscape, we apply gradient clipping, restricting the maximum global norm of the gradients to 0.5.

4.5.2 Optimization Schedule

The network parameters are optimized using the AdamW optimizer Loshchilov and Hutter [2019]. AdamW is chosen specifically over standard Adam because it correctly decouples L2 weight decay (set to 10^{-4}) from the adaptive gradient updates, which is critical for regularizing deep U-Nets and preventing overfitting to the training set. The learning rate is initialized at 3×10^{-4} and decays following a `CosineAnnealingLR` schedule across 100 epochs, allowing for aggressive early exploration and fine-grained late-stage convergence.

5. RESULTS, BENCHMARKING, AND CONCLUSION

5.1 Forward Model Validation

Before deploying the Physics-Informed Neural Network (PINN) for inverse problem solving, it is crucial to rigorously validate the forward analytical solver used to generate the training data. The forward model constructs 3D thermal fields from known nanoparticle arrays and optical excitation parameters.

5.1.1 Benchmark Against Published Data

To ensure physical accuracy, our Green's function-based forward solver was benchmarked against the established thermal profiles of nanoparticle arrays published by Baffou *et al.* Baffou et al. [2013]. A key parameter in defining the thermal distribution of an array of plasmonic nanoparticles is the dimensionless collective heating parameter, ζ_2. It governs the transition from localized heating (where each nanoparticle acts as an isolated hot spot) to collective heating (where the thermal fields of adjacent nanoparticles merge, creating a macroscopic thermal profile).

As the illumination intensity or array density increases, the temperature profile transitions continuously. Our forward solver perfectly replicates these limits. In the low ζ_2 limit, the thermal map resolves individual nanoparticles. At high ζ_2, a broad Gaussian-like temperature envelope dominates the array, seamlessly matching the theoretical models Baffou et al. [2013], Baffou, Quidant, and Girard [2010b].

5.1.2 Multi-Layer System Verification

Practical thermoplasmonic systems are rarely homogeneous; they typically comprise a substrate, a fluid medium, and sometimes a capping superstrate. Our solver incorporates the method of images to rigorously account for thermal boundary resistance and conductivity mismatch across multiple layers. We validated the solver across four distinct topological configurations, defining the thermal conductivities of the substrate (k_1), medium (k_2), and superstrate (k_3):

1. **Homogeneous ($k_1 = k_2 = k_3$):** Modeled entirely in water. The steady-state temperature decays perfectly as $1/r$, exactly matching the analytical solution for a point source in an infinite medium.

2. **Substrate-Supported ($k_1 \neq k_2 = k_3$):** A glass substrate with ethanol medium. The thermal boundary at the glass-ethanol interface causes an asymmetric z-profile due to heat being preferentially conducted into the higher-conductivity glass ($k \approx 1.38$ W/mK) compared to ethanol ($k \approx 0.17$ W/mK).

3. **Superstrate-Covered ($k_1 = k_2 \neq k_3$):** Water medium capped by a PDMS superstrate. The top boundary acts as a thermal barrier, altering the upward diffusion path and flattening the thermal gradient near the boundary.

4. **Heterogeneous Sandwich ($k_1 \neq k_2 \neq k_3$):** A full three-layer complexity comprising a glass substrate, IPA medium, and PDMS superstrate. The solver correctly applies boundary conditions at both interfaces simultaneously, generating complex but physically sound reflections and transmissions of the thermal field.

For each case, 2D temperature maps and cross-sectional profiles were generated and compared against finite-element method (FEM) equivalents, showing negligible relative error (typically $< 0.1\%$).

[PLACEHOLDER FIGURE]

Fig. 5.1: Forward model validation across four topological test cases. From left to right: input phase map, ground truth temperature, PINN prediction, and relative error map. The cross-sectional profiles in the lower panels demonstrate exact agreement between analytical and boundary-adjusted solutions.

5.2 PINN 3D Reconstruction Results

The core capability of our PINN is the ability to map a 2D optical phase image (Quantitative Phase Imaging, QPI) to the full 3D temperature distribution. The network maps $\Delta\phi(x, y) \rightarrow T(x, y, z)$ utilizing a branching architecture.

5.2.1 Qualitative Assessment

Visual inspection of the reconstructed 3D thermal volumes reveals exceptional fidelity. The network accurately predicts the overall morphology of the heat distribution. XY slices taken at various z-planes match the ground truth well, correctly broadening and decreasing in peak intensity as the observation plane moves away from the source plane. XZ and YZ cross-sections accurately capture the depth-dependent thermal diffusion, maintaining the correct boundary asymmetries encoded by the physical topology parameters.

[PLACEHOLDER FIGURE]

Fig. 5.2: Qualitative PINN reconstruction results. Slices at varying z-depths show the input phase map (left), ground truth temperature (middle), and the PINN predicted 3D output (right). Error maps show deviations primarily confined to sub-pixel high-gradient regions.

5.2.2 Z-Axis Source Localization

One of the most challenging aspects of 2D-to-3D inversion is determining the exact depth (z_0) of the heat source. Thermal diffusion rapidly blurs spatial high-frequency components, making the inverse problem ill-posed.

Our architecture leverages a specialized Z-Locator branch employing a spatial softmax operation. This ensures the predicted depth distribution sums to one, effectively acting as a probabilistic weighting. This inductive bias significantly improves z-axis resolution. The model achieves a high z-accuracy, with over 92% of test cases localized to the exact correct z-plane, and a mean absolute error (MAE) in z-index of less than 0.2 pixels. The softmax weighting helps disambiguate depth by preventing the network from predicting diffuse, unphysical source distributions stretched across multiple planes.

[PLACEHOLDER FIGURE]

Fig. 5.3: Z-localization accuracy of the PINN. The softmax-driven Z-locator branch successfully isolates the source plane with minimal error, overcoming the ill-posed nature of depth recovery from 2D blurred phase maps.

5.2.3 Temperature Magnitude Recovery

Alongside spatial distribution, recovering the absolute peak temperature is vital. The Magnitude Branch of the PINN specifically regresses the scaling factor $\Delta T_{\max}$. Overall peak temperature prediction accuracy is consistently within 2-3% of the ground truth. Error analysis reveals that the network slightly under-predicts the temperature in extreme cases—specifically for highly dense, randomly distributed nanoparticle clusters with strong collective heating. In these geometries, the thermal gradient is exceptionally sharp, and the finite resolution of the training grid (1024×1024) acts as a slight low-pass filter, mildly capping the maximum predicted value.

5.2.4 Cross-Cut Profile Analysis

Evaluating 1D cross-cuts along the x, y, and z axes provides a strict test of morphological accuracy. As shown in the horizontal and vertical profiles, the PINN overlay matches the ground truth T-profile tightly across four orders of magnitude (when viewed on a logarithmic scale).

One minor artifact observed is an edge effect near the boundaries of the computational domain. Because convolutional neural networks often employ zero-padding or replicate-padding, the predicted thermal field near the extreme edges of the field-of-view sometimes decays slightly faster than the true $1/r$ analytical tail. However, these edge effects are generally restricted to the outer 5% of the simulation window and do not affect the active region of interest.

[PLACEHOLDER FIGURE]

Fig. 5.4: 1D temperature profiles (cross-cuts) along the x and z axes. The PINN perfectly tracks the $1/r$ thermal decay and respects the asymmetric boundary conditions induced by multi-layer environments.

5.3 Training Convergence Analysis

The training of a PINN balances data-driven empirical loss (MSE between predicted and true temperatures) and physics-driven regularization (adherence to the heat equation).

Monitoring the loss component evolution over epochs demonstrates smooth convergence. The data-driven loss drops rapidly in the first 20 epochs, capturing the low-frequency bulk thermal distribution. The physics loss (specifically the Laplacian constraint $\nabla^2 T = 0$ outside the source) converges more slowly, requiring the network to refine high-frequency spatial gradients. We observed that heavily weighting the Laplacian loss early in training causes instability; however, a graduated weighting strategy allows the physics term to pull the network towards thermodynamically valid solutions without hindering the initial data fitting. The model was trained on an NVIDIA GPU, requiring approximately 45 ms per forward pass, proving its viability for rapid inference.

[PLACEHOLDER FIGURE]

Fig. 5.5: Training convergence curves showing the evolution of the data loss, physics (Laplacian) loss, and Z-locator loss over epochs.

5.4 Experimental Data Processing Results

To test the applicability to real-world measurements, we processed experimental QPI data obtained from a quadriwave lateral shearing interferometer (QWLSI).

The experimental pipeline ingests raw holograms. Through Fourier domain filtering, we isolated the first-order sideband and applied a phase-unwrapping algorithm to yield the cleaned phase map $\Delta\phi(x, y)$. We successfully implemented camera harmonic noise

reduction, isolating and removing structural noise peaks in the Fourier domain. The before/after comparison shows a substantial reduction in background phase ripple, yielding a clean thermal footprint ready for PINN inference.

[PLACEHOLDER FIGURE]

Fig. 5.6: Experimental data processing pipeline. Raw holograms are filtered in the Fourier domain to remove camera harmonics, yielding high signal-to-noise ratio phase maps for thermal reconstruction.

5.5 Discussion

5.5.1 Strengths of the Approach

The primary strength of the PINN approach is that physics constraints prevent unphysical solutions. Even when confronted with noisy or partial phase data, the Laplacian regularization ensures that heat flows from hot to cold without anomalous sinks or sources. Furthermore, the network demonstrates robust generalization. Trained on a diverse distribution of randomized topologies and materials, it can instantly reconstruct thermal fields for novel nanoparticle configurations without requiring time-consuming FEM recalculations. Once trained, the inference time is on the order of milliseconds, enabling real-time 3D thermal monitoring.

5.5.2 Current Limitations

Despite these successes, we must honestly acknowledge several limitations. Most importantly, the synthetic-to-real domain gap remains the biggest open challenge. The network is trained entirely on idealized synthetic data. While our forward model is rigorous, real experimental holograms are plagued by optical aberrations, non-uniform beam profiles, scattering artifacts, and complex sensor noise that we do not perfectly simulate. Bridging this gap is critical for reliable real-world inference. Second, the system relies on a single-frequency assumption (monochromatic probe), limiting the depth resolution inherently constrained by the depth-of-field of a single wavelength. Third, the current model assumes steady-state heating and cannot capture transient thermal dynamics. Finally, the spatial resolution is strictly limited by the training grid size; resolving nanoscale thermal gradients requires prohibitively large memory footprints during 3D convolution.

5.6 Conclusion

In this project, we have developed a robust computational framework for 3D thermoplasmonics. The four main contributions of this work are: 1. Development of a highly

accurate, multi-layer Green's function forward solver for plasmonic heating. 2. Design and training of a bespoke PINN architecture capable of 2D-to-3D thermal inversion. 3. Implementation of a specialized Z-Locator and Magnitude branch to resolve the ill-posed depth and scale problems. 4. Construction of an experimental processing pipeline to bridge QPI measurements with neural network inference.

The key finding of this research is that a properly constrained Physics-Informed Neural Network can successfully and efficiently recover 3D thermal structures from 2D optical phase images, overcoming traditional resolution limits.

5.7 Future Directions

Future work should focus on closing the synthetic-to-real domain gap via transfer learning. Specifically, we plan to fine-tune the synthetic base model using a small, highly calibrated dataset of experimental measurements, such as those obtained from isolated gold nanoparticles with well-understood heating responses.

Additionally, we can address the depth ambiguity by extending the QPI technique to multi-wavelength Thermo-Optic Quantitative Spatial Imaging (TQSI). Using distinct probe wavelengths with different chromatic dispersions would provide independent phase channels, offering the algebraic constraints necessary for robust z-discrimination.

Finally, we intend to expand the physics loss formulation to include the transient time-dependent heat equation, $\rho c_p \frac{\partial T}{\partial t} = \nabla \cdot (\kappa \nabla T) + Q$. By supplying time-resolved phase maps from high-speed pulsed laser experiments, the PINN could be trained to track non-equilibrium nano-thermal dynamics.

Appendices

I Code Snippets: Physics-Informed Loss Function

This appendix provides the complete software implementations, numerical discretization details, physical material property constants, and deep learning training hyperparameters used across the 3D thermoplasmonic temperature field reconstruction pipeline.

I.1 The FastAdaptiveLagrangianLoss Implementation

Solving the 3D thermoplasmonic thermal inverse problem from two-dimensional quantitative phase imaging (QPI) data is inherently ill-posed due to phase integration along the optical axis (z-axis). To regularize the reconstruction process and enforce thermodynamic consistency, we employ the `FastAdaptiveLagrangianLoss` function.

This composite loss function integrates six distinct penalty components:

1. **Domain-weighted L1 data losses** ($\mathcal{L}_{\text{src}}$ and $\mathcal{L}_{\text{bg}}$): Differentiates between high-gradient heat source voxels within plasmonic nanoparticles (weighted by 5.0) and surrounding fluid background voxels (weighted by 2.0).

2. **Boundary condition loss** ($\mathcal{L}_{\text{bc}}$): Penalizes discrepancies along all six spatial bounding faces of the 3D simulation volume to enforce ambient thermal equilibrium at the boundary walls.

3. **Transverse gradient matching** ($\mathcal{L}_{\text{grad}_{xy}}$): Aligns localized spatial derivatives ($\partial T/\partial x, \partial T/\partial y$) between predictions and targets to maintain sharp thermal gradients.

4. **Z-axial localization loss** ($\mathcal{L}_{\text{zloc}}$): Applies 2D adaptive max-pooling (5×5 spatial grid) across axial slice profiles to constrain vertical heat spreading and prevent structural warping.

5. **Physics-informed residual loss** ($\mathcal{L}_{\text{lap}}$): Enforces the steady-state heat conduction equation $\nabla \cdot (\kappa \nabla T) = 0$ in the source-free background fluid domain ($S = 0$) using a differentiable 3D finite-difference convolution operator.

Listing 1 presents the full, self-contained PyTorch implementation of the `FastAdaptiveLagrangianL` module.

```
import torch
import torch.nn as nn
import torch.nn.functional as F

class FastAdaptiveLagrangianLoss(nn.Module):
    """
    Stabilized multi-objective physics-informed loss function for
        3D thermoplasmonics.
```

```
    Combines scale-invariant L1 data fidelity, boundary constraints, spatial gradient
    matching, axial peak localization, and discrete 3D Laplacian heat equation residuals.
    """
    def __init__(self, dx_um, dz_um):
        super().__init__()
        # Calculate non-isotropic grid aspect ratio r_z = (dx / dz)
        self.dz_ratio = (dx_um / dz_um)
        self.dz_ratio_sq = self.dz_ratio ** 2

        # Initialize anisotropic 3D finite-difference Laplacian kernel tensor (1, 1, 3, 3, 3)
        weight = torch.zeros(1, 1, 3, 3, 3)
        # Central node weight: -2 * (2 + r_z^2)
        weight[0, 0, 1, 1, 1] = -2.0 * (2.0 + self.dz_ratio_sq)
        # Transverse neighbor weights (X and Y directions): +1.0
        weight[0, 0, 1, 1, 0] = 1.0; weight[0, 0, 1, 1, 2] = 1.0
        weight[0, 0, 1, 0, 1] = 1.0; weight[0, 0, 1, 2, 1] = 1.0
        # Axial neighbor weights (Z direction): +r_z^2
        weight[0, 0, 0, 1, 1] = self.dz_ratio_sq; weight[0, 0, 2, 1, 1] = self.dz_ratio_sq

        # Register fixed kernel buffer for GPU execution without gradient tracking
        self.register_buffer('lap_kernel', weight)

    def forward(self, T_pred, T_true, k_3d, source_mask):
        # Compute maximum true temperature per sample for scale normalization
        T_max = T_true.amax(dim=(1, 2, 3, 4), keepdim=True) + 1e-5
        p_norm = T_pred / T_max
        t_norm = T_true / T_max

        # Construct binary regional masks for heat source and fluid background
        src_mask_f = (source_mask > 0).float()
        bg_mask_f = (source_mask == 0).float()

```

```
        # Elementwise absolute error tensor
        l1_diff = torch.abs(p_norm - t_norm)

        # 1. Region-weighted data-driving L1 losses
        loss_src = (torch.sum(l1_diff * src_mask_f) / (torch.sum(
            src_mask_f) + 1e-5)) * 5.0
        loss_bg = (torch.sum(l1_diff * bg_mask_f) / (torch.sum(
            bg_mask_f) + 1e-5)) * 2.0

        # 2. Boundary condition loss across all 6 bounding faces
            of the 3D volume
        loss_bc = (F.l1_loss(p_norm[:, :, 0, :, :], t_norm[:, :,
            0, :, :]) +
                   F.l1_loss(p_norm[:, :, -1, :, :], t_norm[:, :,
                       -1, :, :]) +
                   F.l1_loss(p_norm[:, :, :, 0, :], t_norm[:, :,
                      :, 0, :]) +
                   F.l1_loss(p_norm[:, :, :, -1, :], t_norm[:, :,
                       :, -1, :]) +
                   F.l1_loss(p_norm[:, :, :, :, 0], t_norm[:, :,
                      :, :, 0]) +
                   F.l1_loss(p_norm[:, :, :, :, -1], t_norm[:, :,
                       :, :, -1])) * 2.0

        # 3. Transverse spatial finite-difference gradient
            matching (X and Y directions)
        p_dx, t_dx = p_norm[..., 1:] - p_norm[..., :-1], t_norm
            [..., 1:] - t_norm[..., :-1]
        p_dy, t_dy = p_norm[..., 1:, :] - p_norm[..., :-1, :],
            t_norm[..., 1:, :] - t_norm[..., :-1, :]
        loss_grad_xy = (F.l1_loss(p_dx, t_dx) + F.l1_loss(p_dy,
            t_dy)) * 2.0

        # 4. Axial Z-profile peak localization using 2D adaptive
            max-pooling
        B, C, Z, Y, X = p_norm.shape
        p_peaks = F.adaptive_max_pool2d(p_norm.view(B*Z, C, Y, X)
            , (5, 5)).view(B, C, Z, 25).mean(dim=-1)
        t_peaks = F.adaptive_max_pool2d(t_norm.view(B*Z, C, Y, X)
            , (5, 5)).view(B, C, Z, 25).mean(dim=-1)
        loss_zloc = F.l1_loss(p_peaks, t_peaks) * 2.0
```

```
65
66          # 5. Physics-informed steady-state heat equation residual
                via 3D convolution
67          laplacian_field = F.conv3d(T_pred, self.lap_kernel)
68          k_inner = k_3d[:, :, 1:-1, 1:-1, 1:-1]
69          lap_loss_raw = torch.abs(k_inner * laplacian_field) / (
               T_max + 1e-5)
70
71          # Mask out heat sources (S > 0) to enforce laplacian = 0
               strictly in fluid background
72          src_inner = source_mask[:, :, 1:-1, 1:-1, 1:-1]
73          lap_enforcement_mask = (src_inner == 0).float()
74          loss_laplacian = torch.sum(lap_loss_raw *
               lap_enforcement_mask) / (torch.sum(
               lap_enforcement_mask) + 1e-5)
75
76          # Aggregate weighted multi-objective composite loss
77          total_loss = loss_bg + loss_src + loss_bc + loss_grad_xy
               + loss_zloc + (loss_laplacian * 0.05)
78
79          # Extract unscaled scalar metrics for tracking and
               diagnostic logging
80          with torch.no_grad():
81              scalar_t_max = T_max.mean().item()
82              metrics = {
83                  'bg': loss_bg.item(),
84                  'src': loss_src.item(),
85                  'bc': loss_bc.item() * scalar_t_max,
86                  'lap': loss_laplacian.item() * scalar_t_max,
87                  'zloc': loss_zloc.item() * scalar_t_max,
88                  'grad': loss_grad_xy.item() * scalar_t_max
89              }
90
91          return total_loss, metrics
```

Listing 1: PyTorch implementation of the `FastAdaptiveLagrangianLoss` class for physics-informed 3D temperature field inversion.

I.2 The 3D Laplacian Kernel Construction

In stationary thermoplasmonic heat dissipation, the temperature distribution $T(\boldsymbol{r})$ in homogeneous background media follows Laplace's equation:

$$\nabla^2 T(\boldsymbol{r}) = \frac{\partial^2 T}{\partial x^2} + \frac{\partial^2 T}{\partial y^2} + \frac{\partial^2 T}{\partial z^2} = 0. \tag{.1}$$

To evaluate Equation (.1) on a discretized 3D Cartesian grid with non-uniform grid spacing $(\Delta x, \Delta y, \Delta z)$ where $\Delta x = \Delta y \neq \Delta z$, we formulate a 7-point 3D central finite-difference stencil. Expanding second-order derivatives yields:

$$\left.\frac{\partial^2 T}{\partial x^2}\right|_{i,j,k} \approx \frac{T_{i+1,j,k} - 2T_{i,j,k} + T_{i-1,j,k}}{\Delta x^2}, \tag{.2}$$

$$\left.\frac{\partial^2 T}{\partial y^2}\right|_{i,j,k} \approx \frac{T_{i,j+1,k} - 2T_{i,j,k} + T_{i,j-1,k}}{\Delta x^2}, \tag{.3}$$

$$\left.\frac{\partial^2 T}{\partial z^2}\right|_{i,j,k} \approx \frac{T_{i,j,k+1} - 2T_{i,j,k} + T_{i,j,k-1}}{\Delta z^2}. \tag{.4}$$

Multiplying the complete operator $\nabla^2 T_{i,j,k}$ by Δx^2 scales the stencil relative to the transverse sampling distance:

$$\Delta x^2 \nabla^2 T_{i,j,k} = \Big(T_{i+1,j,k} + T_{i-1,j,k} + T_{i,j+1,k} + T_{i,j-1,k} - 4T_{i,j,k}\Big) + r_z^2 \Big(T_{i,j,k+1} - 2T_{i,j,k} + T_{i,j,k-1}\Big), \tag{.5}$$

where $r_z \equiv \Delta x / \Delta z$ is the spatial grid aspect ratio. Combining terms associated with the central grid point (i, j, k) gives the unnormalized stencil coefficient:

$$C_{\text{center}} = -4 - 2r_z^2 = -2\left(2 + r_z^2\right). \tag{.6}$$

This numerical stencil is constructed as a 5D PyTorch tensor of shape $(1, 1, 3, 3, 3)$ representing a single-input, single-output 3D convolution kernel $\mathcal{K}_{z,y,x}$. Table .1 details the exact correspondence between spatial node offsets $(\Delta z, \Delta y, \Delta x)$ and tensor element values.

The PyTorch initialization snippet demonstrates how this numerical operator is instantiated and registered as a persistent, non-trainable parameter:

```
# Step 1: Compute anisotropic grid aspect ratio
dz_ratio = dx_um / dz_um
dz_ratio_sq = dz_ratio ** 2

# Step 2: Instantiate zero-filled 5D kernel tensor (out_c=1, in_c
    =1, depth=3, height=3, width=3)
weight = torch.zeros(1, 1, 3, 3, 3)
```

Tab. .1: Tensor element mapping of the anisotropic 3D Laplacian convolutional kernel $\mathcal{K}$. Index ordering follows PyTorch `conv3d` format `[out_c, in_c, z, y, x]`.

Kernel Slice (z)	**Tensor Indices** (z, y, x)	**Spatial Offset** $(\Delta z, \Delta y, \Delta x)$	**Finite-Difference V**
$z = 0$	$(0, 1, 1)$	$(-1, 0, 0)$	$r_z^2 = (\Delta x/\Delta z)^2$
$z = 1$	$(1, 1, 0)$	$(0, 0, -1)$	$+1.0$
$z = 1$	$(1, 1, 2)$	$(0, 0, +1)$	$+1.0$
$z = 1$	$(1, 0, 1)$	$(0, -1, 0)$	$+1.0$
$z = 1$	$(1, 2, 1)$	$(0, +1, 0)$	$+1.0$
$z = 1$	$(1, 1, 1)$	$(0, 0, 0)$	$-2(2 + r_z^2)$
$z = 2$	$(2, 1, 1)$	$(+1, 0, 0)$	$r_z^2 = (\Delta x/\Delta z)^2$
All others	20 remaining entries	Off-diagonal corners	0.0

```
7
8  # Step 3: Populate 7-point central finite-difference coefficients
9  weight[0, 0, 1, 1, 1] = -2.0 * (2.0 + dz_ratio_sq)  # Center node
10 weight[0, 0, 1, 1, 0] = 1.0; weight[0, 0, 1, 1, 2] = 1.0  # X-
       axis face neighbors
11 weight[0, 0, 1, 0, 1] = 1.0; weight[0, 0, 1, 2, 1] = 1.0  # Y-
       axis face neighbors
12 weight[0, 0, 0, 1, 1] = dz_ratio_sq; weight[0, 0, 2, 1, 1] =
       dz_ratio_sq  # Z-axis face neighbors
13
14 # Step 4: Register as PyTorch buffer to automatically match model
       GPU device and precision
15 self.register_buffer('lap_kernel', weight)
```

Listing 2: Standalone snippet constructing the anisotropic 3D Laplacian kernel tensor.

By utilizing `self.register_buffer`, the kernel tensor is automatically transferred to the active CUDA device alongside model parameters without adding trainable degrees of freedom or incurring memory re-allocation overhead during backward passes.

I.3 Materials Database

Accurate forward modeling and inverse thermal reconstruction depend on exact thermophysical and optical material constants. The phase change $\Delta\phi(x, y)$ detected in Quantitative Phase Imaging (QPI) is directly governed by the first-order thermo-optic coefficient $\beta_1 \equiv \mathrm{d}n/\mathrm{d}T$, which relates local temperature rise ΔT to refractive index variations Δn:

$$\Delta n(x, y, z) = \beta_1 \cdot \Delta T(x, y, z). \tag{.7}$$

Table .2 lists the thermal conductivity κ, thermo-optic coefficient β_1, and complex refractive index $n = n_{\text{real}} + i n_{\text{imag}}$ (at excitation wavelength $\lambda = 532$ nm) for all liquid media, solid substrates, and plasmonic nanostructures stored in the `MATERIALS_DB` simulation engine Baffou et al. [2012, 2020], Johnson and Christy [1972].

Tab. .2: Material properties database (`MATERIALS_DB`) utilized for 3D thermoplasmonic forward simulations and neural network feature conditioning. Optical refractive indices n are specified at illumination wavelength $\lambda = 532$ nm.

Material	**Phase / Class**	**Thermal Conductivity κ** ($\mathrm{W\,m^{-1}\,K^{-1}}$)	**Thermo-Optic Coeff. β_1** (10^{-4} $\mathrm{K^{-1}}$)	**F**
Water (H_2O)	Ambient Liquid	0.60	-1.0514	
Glass (Soda-Lime)	Substrate	1.00	$+0.1000$	
Silica Glass (SiO_2)	Substrate	1.40	$+0.1000$	
Ethanol (C_2H_5OH)	Medium Solvent	0.17	-4.0000	
PDMS	Polymer Matrix	0.15	-4.5000	
IPA (Isopropanol)	Medium Solvent	0.135	-4.5000	
Acetone (C_3H_6O)	Medium Solvent	0.16	-5.0000	
Gold (Au)	Metallic NP	314.00	0.0000	
Silver (Ag)	Metallic NP	429.00	0.0000	

I.4 Training Hyperparameters

The Physics-Guided Inversion UNet (PGI-HNet) architecture was trained on synthetic 3D thermoplasmonic volumes generated across diverse multi-nanoparticle configurations and fluid media. Model optimization was conducted using the AdamW algorithm Loshchilov and Hutter [2019] with Automatic Mixed Precision (AMP) and gradient norm clipping.

Table .3 summarizes all hyperparameters governing network architecture configuration, optimization parameters, dataset volume extents, and physics loss curriculum schedules.

Tab. .3: Complete hyperparameter specification for training the PGI-HNet model.

Hyperparameter Category	Parameter / Setting	Value / Specification
Model Architecture	Base Feature Channels (bf)	16
	Spatial Grid Dimensions	$128 \times 128 \times 32$ ($N_x \times N_y \times N$
	Physical Volume Dimensions	$12.8 \times 12.8 \times 32.0$ μm^3 ($L_x \times$
	Spatial Voxel Resolution	$\Delta x = \Delta y = 0.10$ μm, $\Delta z = 1.$
Optimization	Optimizer	AdamW Loshchilov and Hutt
	Initial Learning Rate (η_0)	1.0×10^{-3}
	Minimum Learning Rate ($\eta_{\min}$)	1.0×10^{-6}
	Learning Rate Scheduler	CosineAnnealingLR ($T_{\max} = 1$
	Weight Decay (λ_w)	1.0×10^{-4}
	Per-step Batch Size	1 volume tensor
	Gradient Accumulation Steps	4 steps (Effective Batch Size =
	Maximum Gradient Norm	5.0 (`clip_grad_norm_`)
	Total Training Epochs	100 epochs
	Numerical Precision	Mixed Precision (FP16 / PyT
Loss Function Weights	Source Region Data Loss Weight	$w_{\text{src}} = 5.0$
	Background Data Loss Weight	$w_{\text{bg}} = 2.0$
	Boundary Condition Weight	$w_{\text{bc}} = 2.0$
	Transverse Gradient Weight	$w_{\text{grad}} = 2.0$
	Z-Localization Weight	$w_{\text{zloc}} = 2.0$
	Laplacian Physics Base Weight	$w_{\text{lap}} = 0.05$
Physics Curriculum Schedule	Warmup Start Epoch	Epoch 5 ($\lambda_{\text{phys}} = 0.00$)
	Warmup End Epoch	Epoch 20 ($\lambda_{\text{phys}} = 0.30$)
	Weight Ramp Profile	Linear ramp: $\lambda_{\text{phys}}(e) = 0.30 \times$

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